Design and Optimization of Specular Reflecting RIS-Aided VLCP System With Low SNR

Qi Wu, Jian Zhang, Rui-jie Duan, Yan-yu Zhang, Gang Xin, and Dun Li

***Abstract*—This paper investigates the optimization of the integrated visible light communication and positioning (VLCP) system performance with the aid of specular reflecting reconfigurable intelligent surface (RIS) when the received signal is subject to low signal-to-noise ration (SNR). Specifically, we focus on the low SNR asymptotic capacity as the criterion of communication indicator whereas the Cramer-Rao low bound (CRLB) as the positioning performance indicator. The VLCP system performance in low SNR is discussed since the scenario may occur if the receiver encounters substantial noise interference (low illumination conditions such as corridors) or the transmitter imposes restrictive limitations on the intensity (such as biology darkroom lab). We derive the CRLB and asymptotic capacity of the RIS-aided multi-LED VLCP system in low SNR and dig out the relationship between the communication or positioning performance and the parameters of RIS. Two optimization problems are formulated with the maximization of the capacity as the optimization object while satisfying the constraints on the CRLB and RIS parameters. To address the non-convex optimization problems, we leverage convex relaxation, the monotonicity of the function, semidefinite relaxation (SDR), and greedy algorithms. Based on these techniques, we propose the integrated two-stage greedy algorithm and greedy strategy-based convex relaxation algorithm to determine the optimal configuration of RIS. From the simulation results, we show that the application of RIS can enhance the VLCP system performance in low SNR obviously, and meanwhile, the proposed algorithms are effective methods for the proposed system performance optimization problems.**

***Index Terms*—Visible light communication, visible light positioning, reconfigurable intelligent surface, low SNR asymptotic capacity, Cramer-Rao lower bound, optimization algorithm.**

I. Introduction

With the increasing demand for high-speed communication and high-accuracy positioning in future indoor smart home network and internet-of-thing (IoT) scenario, integrated visible light communication and positioning (VLCP) technology plays an important role in satisfying the urgent requirements of next-generation wireless systems [1], [2]. Combining the capability of visible light communication (VLC) and visible light positioning (VLP), the application of VLCP technology is willing to achieve the integration of communication, lighting, and positioning [3]. Through the advanced technology, the abundant bandwidth available in the visible light spectrum can be fully used to complement the radio frequency (RF) systems strained by the massive number of users expected in the upcoming sixth generation (6G) communication networks [4]. Meanwhile, the existing light-emitting diodes (LEDs) can be utilized to enable communication and positioning capabilities, which demonstrates the resource efficiency and environmental friendliness of this technology. Compared with traditional RF wireless technologies, VLCP has significant advantages such as no electromagnetic interference, high energy efficiency, high security, and low cost [5].

The current research on VLCP mainly focuses on designing integrated signal transmission schemes to achieve the integration of communication and positioning functions [6], [7], [8], [9]. Moreover, by studying the parameter configuration and resource allocation issues in VLCP systems, the key performance indicators such as positioning accuracy and communication rate are jointly optimized [10], [11], [12], [13]. However, current research is mostly confined to the conventional optimization aspects of signal design and power allocation, failing to fully explore and utilize the spatial degrees of freedom at the physical layer for system performance optimization. Meanwhile, the nanoscale wavelength of visible light renders it highly susceptible to obstructions, shadows, and occlusions, which results in the significant sensitivity of VLCP signals to obstacles and substantial path loss during transmission [14].

In order to overcome the line-of-sight (LoS) blockage problem and dig out the spatial degrees of freedom for VLCP system, the specular reflecting reconfigurable intelligent surface (RIS) composed of a mirror array is considered in our work to aid the implement of VLCP technology. By constructing robust non-line-of-sight (NLoS) communication channels via a dynamically adjusted mirror array, uninterrupted data transmission is ensured even in scenarios where LoS connectivity becomes obstructed [15], and meanwhile, the communication performance is improved [16], [17], [18]. Meanwhile, according to the related researches [19], [20], the intelligent mirrors are proved to cope with the positioning problem when the light transmission from LEDs is inadequate. What's more, the RIS-aided VLCP system is firstly investigated in [21] to focus on the positioning metric while maintaining the corresponding communication performance constraints.

The RIS in visible light spectrum has been widely investigated in the last few years, however, the application of RIS in the

Received 20 May 2025; revised 8 July 2025; accepted 10 July 2025. Date of publication 4 September 2025; date of current version 11 September 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB2802804 and in part by the National Natural Science Foundation of China (NSFC) under Grant 62331024. *(Corresponding author: Jian Zhang.)*

The authors are with Information Engineering University, Henan Province 450000, China (e-mail: qi_wu5757@163.com; Zhang_xinda@126.com; jerryduan0206@outlook.com; xindazhang@126.com; downxg@163.com; idun_xd@163.com).

Digital Object Identifier 10.1109/JPHOT.2025.3589469

scenario when the received signal is subject to low signal-to-ratio (SNR) has not been investigated in current researches. Such scenarios can be applicable to various environments, including indoor corridors, laboratory anechoic chambers, and underground coal mine communications [22]. With the significant capability of addressing LoS blockage, RIS is highly suitable for enhancing signal strength and coverage in low SNR conditions. Consequently, this paper explores a RIS-assisted VLCP system in low SNR condition and investigates the system performance through characterizing the communication metric chosen as asymptotic capacity and positioning indicator expressed as Cramer-Rao low bound (CRLB). To the best of our knowledge, the integrated VLCP system in the presence of a mirror array in low SNR scenario is first studied in the literature. Our key contributions are summarized as follows:

- We investigate a specular reflecting RIS-aided VLCP system in low SNR to achieve simultaneous communication, positioning and illumination, and meanwhile, enhance the quality of received signal and mitigate the LoS blockage effect.
- Based on the proposed system model, the asymptotic capacity in low-SNR regimes and the CRLB for positioning error are analytically derived. Furthermore, according to the shared common term in low SNR capacity results, two optimization problems are formulated with the objective of maximizing the asymptotic capacity, where the RIS assignment matrix serves as the optimization variable under the positioning accuracy constraint.
- To address the non-convex optimization problems, we leverage convex relaxation, the monotonicity of the function, semidefinite relaxation (SDR), and greedy algorithms. Based on these techniques, we propose the integrated two-stage greedy algorithm and greedy strategy-based convex relaxation algorithm to determine the optimal configuration of RIS. Specifically, we decompose the original intractable problems into two stages, namely the location constraint satisfaction stage and communication performance maximization stage, and meanwhile, optimize the two stages respectively. The overall optimization algorithms for the primary problems are summarized and the computational complexity is detailed analyzed.
- Under the limit of positioning accuracy, we validate the asymptotic capacity performance in low SNR for the VLCP system with or without the usage of RIS through simulations. Simulation results indicate that the application of RIS in VLCP system can enhance the system performance in low SNR scenario obviously, and meanwhile, demonstrate the effectiveness of the proposed algorithms in solving the corresponding optimization problems.

The remainder of this paper is organized as follows. The integrated VLCP system model in low SNR is introduced in Section II. In Section III, the asymptotic capacity and CRLB for the system are determined respectively. Section IV formulates the asymptotic capacity maximization problems and concludes the corresponding optimization algorithms. Section V presents our simulations on the efficiency of RIS on enhancing the performance of VLCP system and Section VI concludes the paper.

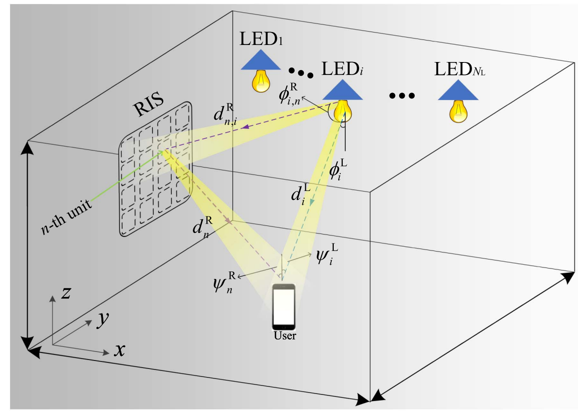


Fig. 1. An integrated VLCP system model with RIS in low SNR.

II. Integrated RIS-Aided VLCP Channel Model

With the capability of communication and positioning, an indoor RIS-aided VLCP system in low SNR scenario is shown in Fig. 1. In the system, the communication and positioning signals are integrated into the transmitter, and meanwhile, the positioning power and communication power are allocated properly to achieve the communication and positioning capability simultaneously [12]. The positioning power refers to the radiant power (or optical power) specifically allocated to the modulation of light signals that carry positional information in a VLP system, whereas the communication power refers to the radiant power (or optical power) allocated to the modulation of light signals that support data communication in a VLC system. The controller in transmitters can adjust the orientation of each mirror to change the transmission environment, provide non-negligible NLoS links, and therefore, increase the corresponding system performance.

Assuming that the set of the LED indexes and reflecting RIS elements indexes are expressed as $\mathcal{L} \triangleq \{1, \ldots, N_L\}$ and $\mathcal{R} \triangleq \{1, \ldots, N_r\}$, respectively. In our work, we focus on one user and the number of photodetector (PD) is assumed to be one. In the provided figure, ${\phi_i}^{\mathrm{L}}$ and ${\phi_i}^{\mathrm{R}}$ denote the irradiance angles, while ${\psi_i}^{\mathrm{L}}$ and ${\psi_n}^{\mathrm{R}}$ represent the incidence angles for LoS links and NLoS links, respectively. ϕ signifies the irradiance angle, and ψ indicates the incidence angle. The index $\forall i \in \mathcal{L}$ corresponds to the i-th LED, whereas $\forall n \in \mathcal{R}$ refers to the n-th RIS element. The notation L is used to denote angles associated with LoS transmission, while R denotes angles related to NLoS transmission. Additionally, ${d_{n,i}}^{\mathrm{R}}$ represents the distance between the i-th LED and the n-th mirror, ${d_n}^{\mathrm{R}}$ denotes the distance from the user to the n-th mirror, and ${d_i}^{\mathrm{L}}$ indicates the LoS transmission distance between the i-th LED and the user. The channel model commonly can be expressed as

$$\boldsymbol{h} = \boldsymbol{h}^{\mathrm{LoS}} + \boldsymbol{h}^{\mathrm{NLoS}}, \tag{1}$$

where $\boldsymbol{h} = [h_1, \ldots, h_{N_L}]^\top \in \mathbb{R}_+^{N_L \times 1}$, $\boldsymbol{h}^{\mathrm{LoS}}$ denotes the LoS channel vector and $\boldsymbol{h}^{\mathrm{NLoS}}$ means the RIS-constructed NLoS channel vector. We focus on the NLoS channel gain formed by the optical transmission link reflected from the mirror array. The NLoS channel gain from diffuse reflecting links is ignored in our work since the negligible effect on the optical transmission. A specular reflecting RIS is deployed on the wall, as this type of RIS demonstrates significantly better performance compared to diffuse reflecting RIS [23]. Specifically, details of the system model are described as follows.

A. LoS Channel Gain

The LoS transmission channel can be modeled as a lambertian model in the proposed VLCP system. The LoS channel gain from the i-th LED can be characterized by [24]

$$h_i^L = \frac{(m+1)A_{re}}{2\pi({d_i}^{\mathrm{L}})^2}\cos^m({\phi_i}^{\mathrm{L}})\cos({\psi_i}^{\mathrm{L}})T_{oc}({\psi_i}^{\mathrm{L}})T_{of}({\psi_i}^{\mathrm{L}}), \tag{2}$$

where m and A_{re} denote the Lambertian index and the area of the PD, respectively. Specifically, m is associated with the LED half-illuminance semi-angle $\Theta_{1/2}$ and can be characterized by $m = -1/\log_2(\cos(\Theta_{1/2}))$. The optical concentrator gain is represented by $T_{oc}(\cdot)$ and $T_{of}(\cdot)$ denotes the optical filter gain. Typically, $T_{of}(\cdot)$ is commonly chosen as a constant whereas $T_{oc}(\cdot)$ is determined by the refractive index within the PD and the field-of-view (FoV). The optical filter gain can be expressed as

$$T_{oc}({\psi_i}^{\mathrm{L}}) = \begin{cases}\beta^2/\sin^2(\Psi_{\mathrm{FoV}}), & 0 \le {\psi_i}^{\mathrm{L}} \le \Psi_{\mathrm{FoV}},\\ 0, & \text{otherwise,}\end{cases} \tag{3}$$

where β is the refractive index within the PD and Ψ_{FoV} represents the FoV. Consequently, the LoS channel vector can be characterized by

$$\boldsymbol{h}^{\mathrm{LoS}} = [h_1^L,\ldots,h_{N_L}^L]^\top; \boldsymbol{h}^{\mathrm{LoS}} \in \mathbb{R}_+^{N_L\times 1}. \tag{4}$$

B. NLoS Channel Gain

From the analysis in [25], the specular reflection-based RIS channel can be described using geometric optics and is modeled as a spherical wave model. Consequently, an extremely near-field condition is determined and an "additive" model is introduced to approximate the near-field propagation characteristics, and accordingly, the "no cross-talk" property determined by extremely near-field condition triggers an assignment matrix to establish the matching relationships among transmitters, intelligent mirrors, and the user. The reflected signal primarily concentrates the power density along the direction specified by the law of reflection, with minimal energy leakage into other directions. From the analysis, if a cascaded LED-RIS-PD channel is set up for a specific LED and an RIS reflecting element by adjusting the coefficients, no other LED can align with this reflecting element and the PD, as dictated by the law of reflection. Specifically, the analysis of the specular reflecting RIS-based NLoS channel gain is as follows.

The NLoS channel gain from the i-th LED to the user via the n-th reflecting mirror can be characterized by [26]

$$h_{i,n}{}^{\mathrm{R}} = \frac{\delta(m+1)A_{re}}{2\pi({d_{n,i}}^{\mathrm{R}}+{d_n}^{\mathrm{R}})^2}\cos^m({\phi_{i,n}}^{\mathrm{R}})\cos({\psi_n}^{\mathrm{R}})T_{oc}({\psi_n}^{\mathrm{R}})T_{of}({\psi_n}^{\mathrm{R}}), \tag{5}$$

where δ is the factor depending on the characteristics of the mirror and usually assumed to be a constant. According to the "no cross-talk" property of the optical RIS channel model, the assignment matrix is proposed based on the assumption that each mirror can only serve one transmitter and the user at a time [25]. Once determining the assignment matrix, the yaw and roll angles of the mirrors can be adjusted using a reverse look-up table to accurately reflect the transmission signal from the transmitters. The service relationship between the transmitters and the mirrors is typically denoted by $\mathbf{G} \triangleq \{\mathbf{g}_1,\ldots,\mathbf{g}_{N_L}\} \in \{0,1\}^{N_r\times N_L}$, where $\mathbf{g}_i = [g_{1,i},\ldots,g_{N_r,i}]^\top$. When a service relationship is established among the i-th transmitter, the n-th mirror and the user, the corresponding element in the assignment matrix is determined (i.e., $g_{n,i} = 1$). Meanwhile, as indicated by the analysis in [27], the narrow beamwidth causes the RIS-reflected energy to become concentrated within a narrow beam when the size of the LED is much smaller than the propagation distance. Based on this, the constraints on the RIS service relationship can be assumed. Given that each mirror can serve at most one transmitter and one user at any given time, the assignment matrix should satisfy the following constraints:

$$g_{n,i} \in \{0,1\}, \quad \forall n \in \mathcal{R}, \tag{6a}$$

$$\sum_{i=1}^{N_L} g_{n,i} \le 1, \quad \forall i \in \mathcal{L}. \tag{6b}$$

Consequently, given the assignment matrix, the channel gain between the i-th transmitter and the user is expressed as

$$h_i^R = \mathbf{g}_i^\top \boldsymbol{h}_i{}^{\mathrm{R}}, \tag{7}$$

where $\boldsymbol{h}_i{}^{\mathrm{R}} \triangleq [h_{i,1}{}^{\mathrm{R}},\ldots,h_{i,N_r}{}^{\mathrm{R}}]^\top \in \mathbb{R}_+^{N_r\times 1}$, and consequently, the NLoS channel vector for the user can be characterized by

$$\boldsymbol{h}^{\mathrm{NLoS}} = [h_1^R,\ldots,h_{N_L}^R]^\top; \boldsymbol{h}^{\mathrm{NLoS}} \in \mathbb{R}_+^{N_L\times 1}. \tag{8}$$

III. Performance Metric

In this section, we consider the asymptotic capacity as the communication metric and the CRLB of the positioning error as the positioning metric. The performance indicators can be derived as follows.

III.

1) Communication Metric: Assuming that the received signal vector $\boldsymbol{y} = (y_1,\ldots,y_{N_L})^\top$ can be derived by the input signal vector $\boldsymbol{x} = (x_1,\ldots,x_{N_L})^\top$ and the channel vector $\boldsymbol{h} = [h_1,\ldots,h_{N_L}]^\top \in \mathbb{R}_+^{N_L\times 1}$, and therefore, characterized by

$$\boldsymbol{y} = \boldsymbol{h}^\top\boldsymbol{x} + \boldsymbol{z}, \tag{9}$$

where $\boldsymbol{z} = (z_1,\ldots,z_{N_L})^\top$ denotes the Gaussian noise vector with zero mean and covariance $\sigma^2\mathbf{1}_{N_L} \in \mathbb{R}_+^{N_L\times 1}$. Considering the peak intensity constraint and sum or individual average intensity constraint given by the intensity-modulation and direct detection (IM/DD) in optical wireless channel (OWC), the asymptotic capacity in low SNR can be characterized differently [22]. Specifically, for the peak and sum average intensity

(PSAI) constraint, the transmit signal should satisfy the following constraints:

$$0 \le x_i \le \mathcal{A},$$
$$\sum_{i=1}^{N_L} \varepsilon_i \le \varepsilon, \forall i \in \mathcal{L}, \tag{10}$$

where $\varepsilon_i = \mathbb{E}[x_i] \in [0, \mathcal{A}]$, $\varepsilon = \alpha\mathcal{A} \in (0, N_L\mathcal{A})$ and $\alpha \in (0, N_L)$. Meanwhile, for the peak and individual average intensity (PIAI) constraint, the transmit signal should satisfy the constraints:

$$0 \le x_i \le \mathcal{A},$$
$$\varepsilon_i \le \varepsilon_{\text{ind}}, \forall i \in \mathcal{L}, \tag{11}$$

where $\varepsilon_{\text{ind}} = \alpha_{\text{ind}}\mathcal{A} \in (0, \mathcal{A})$ and $\alpha_{\text{ind}} \in (0, 1)$.

In our work, we consider the asymptotic capacity defined in two ways for different intensity-constrained scenarios above. Specifically, for the first way, let $\frac{\mathcal{A}}{\sigma} \to 0$, and proportionally, $\frac{\varepsilon}{\sigma} \to 0$ or $\frac{\varepsilon_{\text{ind}}}{\sigma} \to 0$. Another way is achieved through fixing peak SNR $\frac{\mathcal{A}}{\sigma}$, and then, let $\frac{\varepsilon}{\sigma} \to 0$ or $\frac{\varepsilon_{\text{ind}}}{\sigma} \to 0$.

Based on the discussion above, the asymptotic capacity of the VLCP system in low SNR can be characterized as follows [22], [28].

The asymptotic capacity for the PSAI-constrained VLCP system when proportional SNRs $\frac{\mathcal{A}}{\sigma} \to 0$ can be expressed as [22, Theorem 2]

$$C_{[\text{sum}]}(\mathcal{A}, \alpha\mathcal{A}) = \frac{\mathcal{A}}{2\sigma^2}\gamma, \tag{12}$$

where

$$\gamma = \max_{\alpha_i: \sum_{i=1}^{N_L} \alpha_i \le \alpha} \sum_{i=1}^{N_L}\sum_{j=1}^{N_L} h_i h_j \min\{\alpha_i, \alpha_j\} \left(1 - \max\{\alpha_i, \alpha_j\}\right),$$

where $\alpha_i \in [0, 1]$. Particularly, when $\alpha \ge \frac{N_L}{2}$, the asymptotic capacity can be given by [22, Corollary 1]

$$C_{[\text{sum}]}(\mathcal{A}, \alpha\mathcal{A}) = \frac{\mathcal{A}}{8\sigma^2}\|\boldsymbol{h}\|_1^2, \tag{13}$$

where $\|\cdot\|_1$ denotes the ℓ_1-norm of the vector. For fixed peak SNR $\frac{\mathcal{A}}{\sigma}$, and let $\frac{\varepsilon}{\sigma} \to 0$, the asymptotic capacity can be characterized by [22, Theorem 3]

$$C_{[\text{sum}]}(\mathcal{A}, \varepsilon) = \frac{\mathcal{A}\varepsilon}{2\sigma^2 N_L}\|\boldsymbol{h}\|_1^2. \tag{14}$$

As for the PIAI-constrained VLCP system, the asymptotic capacity can be expressed as follows.

When proportional SNRs $\frac{\mathcal{A}}{\sigma} \to 0$, the asymptotic capacity can be expressed as [22, Corollary 2]

$$C_{[\text{ind}]}(\mathcal{A}, \alpha_{\text{ind}}\mathcal{A}) = \frac{\mathcal{A}}{2\sigma^2}\bar{\alpha}_{\text{ind}}(1 - \bar{\alpha}_{\text{ind}})\|\boldsymbol{h}\|_1^2, \tag{15}$$

where $\bar{\alpha}_{\text{ind}} = \min\{\bar{\alpha}_{\text{ind}}, \frac{1}{2}\}$. Meanwhile, for fixed peak SNR and let $\frac{\varepsilon_{\text{ind}}}{\sigma} \to 0$, the asymptotic capacity result is represented by [22, Corollary 3]

$$C_{[\text{ind}]}(\mathcal{A}, \varepsilon_{\text{ind}}) = \frac{\mathcal{A}\varepsilon_{\text{ind}}}{2\sigma^2}\|\boldsymbol{h}\|_1^2. \tag{16}$$

The corresponding proof can refer to [22] and the asymptotic capacity results are summarized in Table I. Consequently, the communication metric can be determined according to the asymptotic capacity result proposed above. Due to the reason that RIS can only influence the channel vector, consequently, we separate the shared term from all capacity results. Specifically, the RIS-determined optimization terms for the asymptotic capacity are γ and $\|\boldsymbol{h}\|_1^2$.

2) Positioning Metric: The positioning metric is chosen as the CRLB of positioning error. Specifically, the received signal strength (RSS) information-based positioning algorithm is adopted in our work to acquire the position of the receiver, given the low cost and high accuracy [29]. Meanwhile, based on the RSS positioning method, the CRLB can be derived as follows.

According to the derivation results in [20], [21], through the maximum likelihood estimator for the location of receiver, the Fisher information matrix (FIM) [30] can be expressed as

$$\mathbf{J}(\mathbf{u}) = \sum_{i=1}^{N_L} \frac{P_{p,i}^2}{\sigma^2}\frac{\partial h_i(\mathbf{u})}{\partial \mathbf{u}}\left(\frac{\partial h_i(\mathbf{u})}{\partial \mathbf{u}}\right)^\top, \tag{17}$$

where $P_{p,i}$ is the positioning power for the i-th LED, $\mathbf{u} = [x_u, y_u, z_u]^\top$ denotes the 3D position of the receiver, where

$$h_i(\mathbf{u}) = h_i^L(\mathbf{u}) + h_i^R(\mathbf{u}), \tag{18}$$

and meanwhile,

$$\frac{\partial h_i(\mathbf{u})}{\partial \mathbf{u}} = \frac{\partial h_i^L(\mathbf{u})}{\partial \mathbf{u}} + \frac{\partial h_i^R(\mathbf{u})}{\partial \mathbf{u}}. \tag{19}$$

The partial derivatives of the LoS channel vector can be calculated directly whereas that related to the NLoS channel vector should be satisfied with:

$$\frac{\partial \boldsymbol{h}^{\text{NLoS}}}{\partial \mathbf{u}_k} = \text{diag}(\frac{\partial \mathbf{H}^{\text{R}}}{\partial \mathbf{u}_k})^\top \text{vec}(\mathbf{G}), \tag{20}$$

where $\text{diag}(\cdot)$ and $\text{vec}(\cdot)$ denote the diagonalization and vectorization operations, $\mathbf{H}^{\text{R}} \triangleq [\boldsymbol{h}_1{}^{\text{R}}, \ldots, \boldsymbol{h}_{N_r}{}^{\text{R}}] \in \mathbb{R}_+^{N_r \times N_L}$. Specifically, assuming that the LEDs are oriented downward and the receiver is oriented upward, the partial derivatives can be calculated as [21]

$$\frac{\partial h_i^L(\mathbf{u})}{\partial x_u} = \frac{-\beta(m+3)(z_{l_i} - z_u)^{m+1}(x_u - x_{l_i})}{||\mathbf{l}_i - \mathbf{u}||_2^{m+5}},$$
$$\frac{\partial h_i^L(\mathbf{u})}{\partial y_u} = \frac{-\beta(m+3)(z_{l_i} - z_u)^{m+1}(y_u - y_{l_i})}{||\mathbf{l}_i - \mathbf{u}||_2^{m+5}},$$
$$\frac{\partial h_i^L(\mathbf{u})}{\partial z_u} = \frac{-\beta(m+1)(z_{l_i} - z_u)^{m}}{||\mathbf{l}_i - \mathbf{u}||_2^{m+3}} + \frac{\beta(m+3)(z_{l_i} - z_u)^{m+2}}{||\mathbf{l}_i - \mathbf{u}||_2^{m+5}},$$
$$\frac{\partial h_{n,i}{}^{\text{R}}(\mathbf{u})}{\partial x_u} = \frac{\zeta(z_{l_i}-\tilde{z}_n)^m(\tilde{z}_n-z_u)(\tilde{x}_n-x_u)}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2} \left\{\frac{||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n - \mathbf{u}||_2} + 3\frac{||\tilde{\mathbf{l}}_n - \mathbf{u}||_2}{||\mathbf{l}_i - \tilde{\mathbf{l}}_n||_2} + 4\right\},$$
$$\frac{\partial h_{n,i}{}^{\text{R}}(\mathbf{u})}{\partial y_u} = \frac{\zeta(z_{l_i} - \tilde{z}_n)^m(\tilde{z}_n - z_u)(\tilde{y}_n - y_u)}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2}$$

TABLE I
SUMMARY OF RIS-AIDED VLCP SYSTEM ASYMPTOTIC CAPACITY IN LOW SNR REGIME

Case	**Peak and sum average intensity constraint**		**Peak and individual average intensity constraint**	
Intensity constraint	Proportionally SNRs	Fixed peak SNR	Proportionally SNRs	Fixed peak SNR
Capacity results	$C_{[\text{sum}]}(\mathcal{A},\alpha\mathcal{A})$	$C_{[\text{sum}]}(\mathcal{A},\varepsilon)$	$C_{[\text{ind}]}(\mathcal{A},\alpha_{\text{ind}}\mathcal{A})$	$C_{[\text{ind}]}(\mathcal{A},\varepsilon_{\text{ind}})$
Shared common term	γ	$\|\boldsymbol{h}\|_1^2$		

$$\left\{\frac{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}+3\frac{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}+4\right\},$$

$$\frac{\partial h_{n,i}{}^{\mathrm{R}}(\mathbf{u})}{\partial z_u}=\frac{\zeta(z_{l_i}-\tilde{z}_n)^m(\tilde{z}_n-z_u)^2}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^4||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m-1}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2^2}\left\{\frac{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}+3\frac{||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}{||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2}+4\right\}-\frac{\zeta(z_{l_i}-\tilde{z}_n)^m}{(||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2+||\tilde{\mathbf{l}}_n-\mathbf{u}||_2)^2||\mathbf{l}_i-\tilde{\mathbf{l}}_n||_2^{m}||\tilde{\mathbf{l}}_n-\mathbf{u}||_2}. \tag{21}$$

where $\mathbf{l}_i \triangleq [x_{l_i}, y_{l_i}, z_{l_i}]$ represents the location of the i-th LED, $\tilde{\mathbf{l}}_n \triangleq [\tilde{x}_n, \tilde{y}_n, \tilde{z}_n]$ denotes the location of the n-th mirror, $\beta = \frac{(m+1)A_{re}T_{oc}(\psi_i{}^{\mathrm{L}})T_{of}(\psi_i{}^{\mathrm{L}})}{2\pi}$ and $\zeta = \frac{\delta(m+1)A_{re}T_{oc}(\psi_n{}^{\mathrm{R}})T_{of}(\psi_n{}^{\mathrm{R}})}{2\pi}$. Combined with (17)–(21), the CRLB of the positioning error can be calculated and specified by $\mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))$.

IV. Asymptotic Capacity Maximization Under Positioning Metric Constraint

This section formulates asymptotic capacity maximization problems with the constraints on positioning metric chosen as CRLB. The elements of the assignment matrix are optimized to maximize the asymptotic capacity.

A. Problem Formulation

As shown in Table I, the asymptotic capacity results in four cases are considered in our work. It is noted that the capacity expression is determined by the shared common term $\|\boldsymbol{h}\|_1^2$ for three cases, and particularly, γ for the asymptotic capacity for the PSAI-constrained VLCP system when proportional SNRs tends to zero. Consequently, the asymptotic capacity maximization problem can be simplified as

$$P\text{:}\quad \max_{\mathbf{G}}\quad \gamma \quad \text{or} \quad \|\boldsymbol{h}\|_1^2 \tag{22a}$$

$$s.t.\quad \mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))\le D_{\text{th}}, \tag{22b}$$

$$g_{n,i}\in\{0,1\},\quad \forall n\in\mathcal{R}, \tag{22c}$$

$$\sum_{i=1}^{N_L} g_{n,i}\le 1,\quad \forall i\in\mathcal{L}. \tag{22d}$$

where D_{th} denotes the threshold of the positioning accuracy and (22b) represents the constraints on the CRLB. Meanwhile, (22c) and (22d) guarantee the constraints on the assignment matrix. Since the existence of the non-convex constraints (such as the integer constraint in (22c), the non-convex constraint in (22b)), the asymptotic capacity maximization problem is a non-convex optimization problem which cannot be solved in polynomial time. As for different optimization objections, the optimization problem can be summarized as two subproblems characterized by

$$\begin{aligned} P1\text{:}\quad &\max_{\mathbf{G}} \quad \|\boldsymbol{h}\|_1 \\ &s.t.\quad \mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))\le D_{\text{th}}, \\ &\qquad g_{n,i}\in\{0,1\},\quad \forall n\in\mathcal{R}, \\ &\qquad \sum_{i=1}^{N_L} g_{n,i}\le 1,\quad \forall i\in\mathcal{L}. \end{aligned} \tag{23}$$

$$\begin{aligned} P2\text{:}\quad &\max_{\mathbf{G}} \quad \gamma \\ &s.t.\quad \mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))\le D_{\text{th}}, \\ &\qquad g_{n,i}\in\{0,1\},\quad \forall n\in\mathcal{R}, \\ &\qquad \sum_{i=1}^{N_L} g_{n,i}\le 1,\quad \forall i\in\mathcal{L}. \end{aligned} \tag{24}$$

The simplification in the optimization object in *P1* is since the non-negativity of the channel gain in VLCP systems. To address the complex non-convex optimization problem, the necessary mathematical processing and optimization algorithm is introduced to tackle the proposed optimization problem.

B. Proposed Algorithm

As for the two proposed optimization problems, we propose two algorithms to solve the proposed problems. Details are summarized as follows.

1) Two-Stage Greedy Algorithm for P1: We consider adopting the two-stage greedy algorithm to solve the optimization problem *P1*. As for the communication performance metric $\|\boldsymbol{h}\|_1$ chosen as the objective function, the variable $\mathbf{G}$ is adjusted to determine $\boldsymbol{h}^{\text{NLoS}}$, and therefore, influence the total channel gain $\boldsymbol{h}$. By activating additional reflection paths, which corresponds to an increase in the number of non-zero elements within the assignment matrix $\mathbf{G}$, the norm of the channel gain $\|\boldsymbol{h}\|_1$ can be effectively enhanced. Meanwhile, according to the relationship between the CRLB and $\mathbf{G}$ characterized by

$$\frac{\partial h_i(\mathbf{u})}{\partial \mathbf{u}}=\frac{\partial h_i^L(\mathbf{u})}{\partial \mathbf{u}}+\mathbf{g}_i^{\top}\frac{\partial \boldsymbol{h}_i{}^{\mathrm{R}}(\mathbf{u})}{\partial \mathbf{u}}, \tag{25}$$

activating the reflection paths with high derivative gain (where the corresponding elements in $\mathbf{g}_i$ are 1) can enhance the positive definiteness of $\mathbf{J}(\mathbf{u})$, thereby reducing $\mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))$. This is since that activating multiple high-order derivative reflection paths leads to an increase in the rank and eigenvalues of $\mathbf{J}(\mathbf{u})$, thereby enhancing the positive definiteness, and therefore, reducing the magnitude of eigenvalues for $\mathbf{J}^{-1}(\mathbf{u})$.

However, the maximization for $\|\boldsymbol{h}\|_1$ typically necessitates the activation of a greater number of reflection paths. However, this approach may inadvertently introduce paths with low derivative gain, which can lead to insufficient information content in $\mathbf{J}(\mathbf{u})$ and consequently result in an increase in $\mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))$. Meanwhile, minimizing $\mathrm{Tr}(\mathbf{J}^{-1}(\mathbf{u}))$ requires prioritizing the selection of reflection paths with high derivative gain, which may compromise the value $\|\boldsymbol{h}\|_1$. For the proposed *P1*, the local optimal choices of reflection surfaces can lead to a global optimum, which satisfying the greedy selection property. Meanwhile, the communication optimization under positioning constraints is independent on previous selections, which exhibiting an optimal substructure. These properties align with the conditions required for the application of greedy algorithms. Consequently, in order to relieve the contradiction between communication and positioning, the two-stage greedy algorithm is adopted in our work to solve problem *P1*.

The greedy algorithm is a heuristic optimization approach that relies on making locally optimal decisions at each step, with the overarching objective of converging towards a globally optimal solution [31]. The adoption of greedy algorithm in this problem can be summarized as two stages: (i) localization constraint satisfaction; (ii) communication performance maximization. As for the first stage, the greedy strategy is to activate the least mirrors to satisfy the positioning accuracy constraint quickly. For the second stage, we adopt the greedy strategy by optimizing the remainder mirrors to maximize the communication metric.

First of all, the assignment matrix $\mathbf{G} \triangleq \mathbf{0}_{N_r \times N_L}$, the FIM matrix $\mathbf{J}(\mathbf{u}) \triangleq \sum_{i=1}^{N_L} (\frac{\partial h_i^L}{\partial \mathbf{u}})(\frac{\partial h_i^L}{\partial \mathbf{u}})^\top$, the sets of the mirrors $\mathcal{R} \triangleq \{1, \ldots, N_r\}$ are initialized. The sensitivity to the receiver position of each candidate pair (n, i) can be calculated by

$$S_{n,i} = \left\| \frac{\partial h_i^L}{\partial \mathbf{u}} + \frac{\partial h_{n,i}{}^{\mathrm{R}}}{\partial \mathbf{u}} \right\|_2^2, \tag{26}$$

where $\|\cdot\|_2$ denotes the ℓ_2-norm of the vector. According to the greedy strategy, the pair (n^*, i^*) contributes the maximum value of the sensitivity from all candidate pairs is chosen, and accordingly, the assignment matrix can be updated in accordance with following principles:

$$\begin{aligned} &G(n^*, i^*) = 1, \\ &\mathcal{R} \leftarrow \mathcal{R} \setminus \{n^*\}. \end{aligned} \tag{27}$$

Assuming it is in the t-th iteration, the FIM matrix $\mathbf{J}^{(t)} = \sum_{i=1}^{N_L} (\frac{\partial h_i^{(t)}}{\partial \mathbf{u}})(\frac{\partial h_i^{(t)}}{\partial \mathbf{u}})^\top$. Based on the updating process above, the FIM matrix in the $(t+1)$-th iteration can be characterized by

$$\mathbf{J}^{(t+1)} = \mathbf{J}^{(t)} + \frac{P_{p,i}^2}{\sigma^2} \Bigg(\left(\frac{\partial h_{i^*}^{(t)}}{\partial \mathbf{u}} \right) \left(\frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}} \right)^\top + \left(\frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}} \right) \left(\frac{\partial h_{i^*}^{(t)}}{\partial \mathbf{u}} \right)^\top + \left(\frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}} \right) \left(\frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}} \right)^\top \Bigg), \tag{28}$$

Algorithm 1: Integrated Two-Stage Greedy Algorithm for **P1**.

Input : N_r, N_L: Number of elements and links;
$\boldsymbol{h}^{\mathrm{LoS}}, \mathbf{H}^{\mathrm{R}}$: Channel components;
$\nabla \mathbf{H}^{\mathrm{R}}$: Gradient tensor;
D_{th}: CRLB threshold;
Output: $\mathbf{G}^*$: Integrated configuration matrix;
Stage 1: Localization Constraint Satisfaction
Initialize localization matrix: $\mathbf{G}_{\mathrm{loc}} \leftarrow \mathbf{0}_{N_r \times N_L}$
Active mirrors: $\mathcal{R} \leftarrow \{1, \ldots, N_r\}$
Initialize FIM: $\mathbf{J} \leftarrow \sum_{i=1}^{N_L} \left(\frac{\partial h_i^{\mathrm{L}}}{\partial \mathbf{u}} \right) \left(\frac{\partial h_i^{\mathrm{L}}}{\partial \mathbf{u}} \right)^\top$
while CRLB $> D_{\mathrm{th}}$ ***and*** $\mathcal{R} \neq \emptyset$ **do**
  $\mathbf{S} \leftarrow \mathbf{0}_{N_r \times N_L}$
  foreach $n \in \mathcal{R}$ **do**
    for $i \leftarrow 1$ **to** N_L **do**
      $S_{n,i} \leftarrow \left\| \frac{\partial h_i^{\mathrm{L}}}{\partial \mathbf{u}} + \frac{\partial h_{n,i}^{\mathrm{R}}}{\partial \mathbf{u}} \right\|_2^2$
    end
  end
  Select optimal pair: $(n^*, i^*) \leftarrow \arg\max S_{n,i}$
  Update configuration: $\mathbf{G}_{\mathrm{loc}}[n^*, i^*] \leftarrow 1$
  Remove used mirror: $\mathcal{R} \leftarrow \mathcal{R} \setminus \{n^*\}$
  Update FIM via equation (28)
  Update derivatives: $\frac{\partial h_{i^*}}{\partial \mathbf{u}} \leftarrow \frac{\partial h_{i^*}}{\partial \mathbf{u}} + \frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}}$
  Update CRLB: CRLB $\leftarrow \mathrm{Tr}(\mathbf{J}^{-1})$
end
Stage 2: Communication Performance Maximization Remaining mirrors: $\mathcal{R}^* \leftarrow \mathcal{R}$
Initialize communication matrix: $\mathbf{G}_{\mathrm{comm}} \leftarrow \mathbf{0}_{N_r \times N_L}$
foreach *mirror* $n \in \mathcal{R}^*$ **do**
  Find optimal link: $i^* \leftarrow \arg\max_{1 \le i \le N_L} h_{n,i}^{\mathrm{R}}$
  Update $\mathbf{G}_{\mathrm{comm}}[n, i^*] \leftarrow 1$
end
Configuration Fusion
$\mathbf{G}^* \leftarrow \mathbf{G}_{\mathrm{loc}} \oplus \mathbf{G}_{\mathrm{comm}}$ where $\oplus$ denotes element-wise OR operation
return $\mathbf{G}^*$

where $h_i^{(t)}$ denotes the i-th channel gain at the t-th iteration, and specifically, $h_i^{(0)} = h_i^L$. The channel gain can be updated based on

$$\frac{\partial h_{i^*}^{(t+1)}}{\partial \mathbf{u}} = \frac{\partial h_{i^*}^{(t)}}{\partial \mathbf{u}} + \frac{\partial h_{n^*,i^*}^{\mathrm{R}}}{\partial \mathbf{u}}. \tag{29}$$

Based on (27) and (28), the assignment matrix can be updated while satisfying the constraints in (22c) and (22d). The updating process is terminated until the constraint in (22b) is satisfied or $\mathcal{R}$ is empty (the positioning accuracy constraint cannot be satisfied even if all the mirrors are used).

As for the communication performance maximization stage, the remainder mirrors are used to maximize the ℓ_1-norm of the channel vector. Specifically, the updated sets of the mirrors are

Algorithm 2: Greedy Strategy-Based Convex Relaxation Algorithm for **P2**.

Input : N_r, N_L: Number of elements and links;
$\boldsymbol{h}^{\mathrm{LoS}}, \mathbf{H}^{\mathrm{R}}$: Channel components;
$\nabla\mathbf{H}^{\mathrm{R}}$: Gradient tensor;
D_{th}: CRLB threshold;
$T_{\max}$: Maximum iterations;
ϵ: Convergence threshold
Output: $\mathbf{G}^*$: Integrated configuration matrix;
Stage 1: Localization Constraint Satisfaction
The same implementation as Algorithm 1
Stage 2: Communication Performance Maximization
Initialize communication matrix: $\mathbf{G}_{\mathrm{comm}} \leftarrow \mathbf{0}_{N_r \times N_L}$
Remaining mirrors: $\mathcal{R}^* \leftarrow \mathcal{R}$
Initialize previous solution: $\mathbf{Q}^{\mathrm{prev}} \leftarrow \mathbf{0}$
Initialize iteration counter: $t \leftarrow 1$
while $t \le T_{\max}$ and $\|\mathbf{Q} - \mathbf{Q}^{prev}\|_F \ge \epsilon$ **do**
  Solve optimal parameters $\mathbf{Q}$ via convex optimization:
   $\mathbf{Q} \leftarrow \arg\max\ f(\mathbf{G}_{\mathrm{comm}})$ (Update with current $\mathbf{G}_{\mathrm{comm}}$)
  Obtain SDP solution $\mathbf{H}^*$: $\mathbf{H}^* \leftarrow \arg\max$ **P2-c** (Using updated $\mathbf{Q}$)
  Generate candidate vector $\boldsymbol{h}^*$ via Gaussian randomization
  Compute relaxed matrix $\tilde{\mathbf{G}}$ via matrix reconstruction
  foreach *mirror* $n \in \mathcal{R}^*$ **do**
   Find optimal link: $i^* \leftarrow \arg\max_{1\le i\le N_L} \tilde{g}_{n,i}$
   Configure single-active mode:
    $\mathbf{G}_{\mathrm{comm}}[n, i^*] \leftarrow 1$
   foreach $i \ne i^*$ **do**
    $\mathbf{G}_{\mathrm{comm}}[n, i] \leftarrow 0$
   end
  end
  Update previous solution: $\mathbf{Q}^{\mathrm{prev}} \leftarrow \mathbf{Q}$
  Update iteration counter: $t \leftarrow t + 1$
end
Configuration Fusion
$\mathbf{G}^* \leftarrow \mathbf{G}_{\mathrm{loc}} \oplus \mathbf{G}_{\mathrm{comm}}$ where $\oplus$ denotes element-wise OR
return $\mathbf{G}^*$

donated as $\mathcal{R}^*$. From (23), we can observe that the objective function is linear with respect to $\mathbf{G}$. Obviously, $\|\boldsymbol{h}\|_1$ increases monotonically with $g_{n,i}$. The constraint in (22d) can be rewritten as

$$\sum_{i=1}^{N_L} g_{n,i} = 1, \quad \forall i \in \mathcal{L}, \quad \forall n \in \mathcal{R}^*. \tag{30}$$

The optimization problem *P1* can be rewritten as

$$\textit{P1-a:} \quad \max_{\mathbf{G}} \mathbf{1}^\top \boldsymbol{h} \tag{31a}$$

Algorithm 3: Exhaustive Search-based Global Configuration Optimization Algorithm.

Input : N_r, N_L: Number of elements and links;
$\boldsymbol{h}^{\mathrm{LoS}}, \mathbf{H}^{\mathrm{R}}$: Channel components;
$\nabla\mathbf{H}^{\mathrm{R}}$: Gradient tensor;
D_{th}: CRLB threshold;
$\mathcal{F}(\mathbf{G})$: Objective function to maximize
Output: $\mathbf{G}^*$: Optimal configuration matrix;
Stage 1: Localization Constraint Satisfaction
Implementation identical to Stage 1 of Algorithm 1, yielding $\mathbf{G}_{\mathrm{loc}}$
Stage 2: Exhaustive Search-Based Communication Optimization
Initialize communication matrix: $\mathbf{G}_{\mathrm{comm}} \leftarrow \mathbf{0}_{N_r \times N_L}$
Remaining mirrors: $\mathcal{R}^* \leftarrow \mathcal{R}$
Initialize candidate set: $\mathcal{S} \leftarrow \{\}$
Initialize optimal value: $f^* \leftarrow -\infty$
foreach *mirror* $n \in \mathcal{R}^*$ **do**
  $\mathcal{S}_{\mathrm{new}} \leftarrow \{\}$
  foreach $\mathbf{G} \in \mathcal{S}$ **do**
   for $i \leftarrow 1$ **to** N_L **do**
    $\mathbf{G}_{\mathrm{new}} \leftarrow \mathbf{G}$
    Append row $\mathbf{e}_i$ (1 at column i)
    $\mathcal{S}_{\mathrm{new}} \leftarrow \mathcal{S}_{\mathrm{new}} \cup \{\mathbf{G}_{\mathrm{new}}\}$
   end
  end
  $\mathcal{S} \leftarrow \mathcal{S}_{\mathrm{new}}$
end
foreach $\mathbf{G} \in \mathcal{S}$ **do**
  $f \leftarrow \mathcal{F}(\mathbf{G})$
  if $f > f^*$ **then**
   $f^* \leftarrow f$
   $\mathbf{G}_{\mathrm{comm}} \leftarrow \mathbf{G}$
  end
end
Configuration Fusion
$\mathbf{G}^* \leftarrow \mathbf{G}_{\mathrm{loc}} \oplus \mathbf{G}_{\mathrm{comm}}$ where $\oplus$ denotes element-wise OR
return $\mathbf{G}^*$

$$s.t. \quad \sum_{i=1}^{N_L} g_{n,i} = 1, \quad \forall i \in \mathcal{L}, \tag{31b}$$

$$g_{n,i} \ge 0, \quad \forall n \in \mathcal{R}^*, \tag{31c}$$

where $\mathbf{1}$ represents the all-one matrix, the objective function can be analyzed as follows.

$$\begin{aligned}\mathbf{1}^\top \boldsymbol{h} &= \mathbf{1}^\top \boldsymbol{h}^{\mathrm{LoS}} + \mathbf{1}^\top \mathrm{diag}(\mathbf{H}^{\mathrm{R}})^\top \mathrm{vec}(\mathbf{G}) \\ &= \sum_{n=1}^{N_r^*} \sum_{i=1}^{N_L} {h_{n,i}}^{\mathrm{R}} g_{n,i} + \mathbf{1}^\top \boldsymbol{h}^{\mathrm{LoS}} \\ &\le \sum_{n=1}^{N_r^*} \max_i {h_{n,i}}^{\mathrm{R}} + \mathbf{1}^\top \boldsymbol{h}^{\mathrm{LoS}},\end{aligned}$$

where N_r^* is the number of the rest mirrors after the localization constraints satisfaction stage, and the equality holds when the index of LED is chosen based on the principle achieving the maximization of $h_{n,i}{}^{\mathrm{R}}$. In other words, the index of LED is determined as

$$\tilde{i} = \arg\max_{i} h_{n,i}{}^{\mathrm{R}}, \quad \forall n \in \mathcal{R}^*. \tag{32}$$

Consequently, based on the kind of chosen principle, the n-th row of the chosen assignment matrix $\tilde{\mathbf{G}}$ is determined by

$$\tilde{g}_{n,i} = \begin{cases} 1, & \text{if } i = \tilde{i}, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

It is observed that $\tilde{\mathbf{G}}$ is subject to the constraint in (22c) and (22d). Meanwhile, the chosen assignment matrix can achieve the maximization of the objective function of the problem.

Based on the two-stage greedy algorithm, the assignment matrix can be fast determined to achieve the maximization of the communication performance metric $\|\boldsymbol{h}\|_1$ under the constraint on the positioning accuracy. Details of the algorithm are summarized in Algorithm 1.

2) Greedy Strategy-Based Convex Relaxation Algorithm for P2: For the optimization of **P2**, the optimization function can be rewritten as

$$\gamma = \max_{\boldsymbol{\alpha}:, \mathbf{1}^\top \boldsymbol{\alpha} \leq \alpha} \left[\boldsymbol{h}^\top \left(\mathbf{M}(\boldsymbol{\alpha}) - \boldsymbol{\alpha}\boldsymbol{\alpha}^\top \right) \boldsymbol{h} \right], \tag{34}$$

where $\mathbf{M}(\boldsymbol{\alpha})$ is the symmetric matrix and the elements of the matrix are determined as $M_{i,j} = \min\{\alpha_i, \alpha_j\}$, $\boldsymbol{\alpha} \triangleq [\alpha_1, \ldots, \alpha_{N_L}]^\top$ is the optimization variable. Note that the Hessian matrix of the objective function is derived as $\nabla^2\gamma = -2\boldsymbol{h}\boldsymbol{h}^\top$, which is negative semidefinite. Consequently, the optimization process for $\boldsymbol{\alpha}$ can be solved through the standard convex optimization technology such as CVX toolbox in MATLAB [32]. After determining the optimal variable $\boldsymbol{\alpha}^*$, the optimization goal in **P2** can be reformulated as

$$\gamma = \boldsymbol{h}^\top \mathbf{Q} \boldsymbol{h}, \tag{35}$$

where $\mathbf{Q} = \mathbf{M}(\boldsymbol{\alpha}^*) - \boldsymbol{\alpha}^* \boldsymbol{\alpha}^{*\top}$ and can be considered as a known symmetric matrix. In order to satisfy the location accuracy constraint, for the first stage, the principle of the element of $\mathbf{G}$ is the same as that in Section IV-B1. Similarly, the optimized assignment matrix element for the communication process is based on the rest mirrors. Consequently, the optimization problem of **P2** for the communication process can be rewritten as

$$\begin{aligned} \textit{P2-a:} \quad & \max_{\mathbf{G}} \quad \boldsymbol{h}^\top \mathbf{Q} \boldsymbol{h} \\ & s.t. \quad g_{n,i} \in \{0,1\}, \quad \forall n \in \mathcal{R}^*, \\ & \qquad \sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall i \in \mathcal{L}. \end{aligned} \tag{36}$$

Proposition 1: It is observed that the matrix $\mathbf{Q}$ is a positive semidefinite matrix when $\alpha_i \in [0,1], \forall i \in \mathcal{L}$.

Proof: Proposition 1 can be proved by constructing the random variable. The expressions and detailed proof are given in Appendix A. □

Based on the positive semi-definiteness of $\mathbf{Q}$, the binary constraints can be relaxed to $g_{n,i} \in [0,1]$. Consequently, the objective problem **P2-a** can be solved using convex optimization methods, such as semidefinite programming. Subsequently, the integer solution can be recovered via random rounding or branch-and-bound techniques. Details of the algorithm are summarized as follows.

Firstly, let $\mathbf{H} = \boldsymbol{h}\boldsymbol{h}^\top$, and the objective function $\boldsymbol{h}^\top \mathbf{Q} \boldsymbol{h}$ can be rewritten as $\mathrm{Tr}(\mathbf{QH})$. Additionally, the semidefinite constraint $\mathbf{H} \succeq 0$ is imposed. Then, the binary variables $g_{n,i} \in \{0,1\}$ is relaxed as continuous variables $g_{n,i} \in [0,1]$, the rank 1 constraint $\mathrm{rank}(\mathbf{H}) = 1$ is ignored, while the allocation constraint $\sum_{i=1}^{N_L} g_{n,i} \leq 1$ is retained. Based on the relaxation, the optimization model is expressed as

$$\begin{aligned} \textit{P2-b:} \max_{\mathbf{G},\mathbf{H}} \quad & \mathrm{Tr}(\mathbf{QH}) \\ \text{s.t.} \quad & \mathbf{H} \succeq \mathbf{0}, \\ & 0 \leq g_{n,i} \leq 1, \quad \forall n, i, \\ & \sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall n. \end{aligned} \tag{37}$$

Note that the relationship between $\boldsymbol{h}$ and $\mathbf{G}$ can be characterized by (1), (4), (7) and (8). The constraints on the assignment matrix can be relaxed into the constraints on $\mathbf{H}$, and the relationship can be characterized by

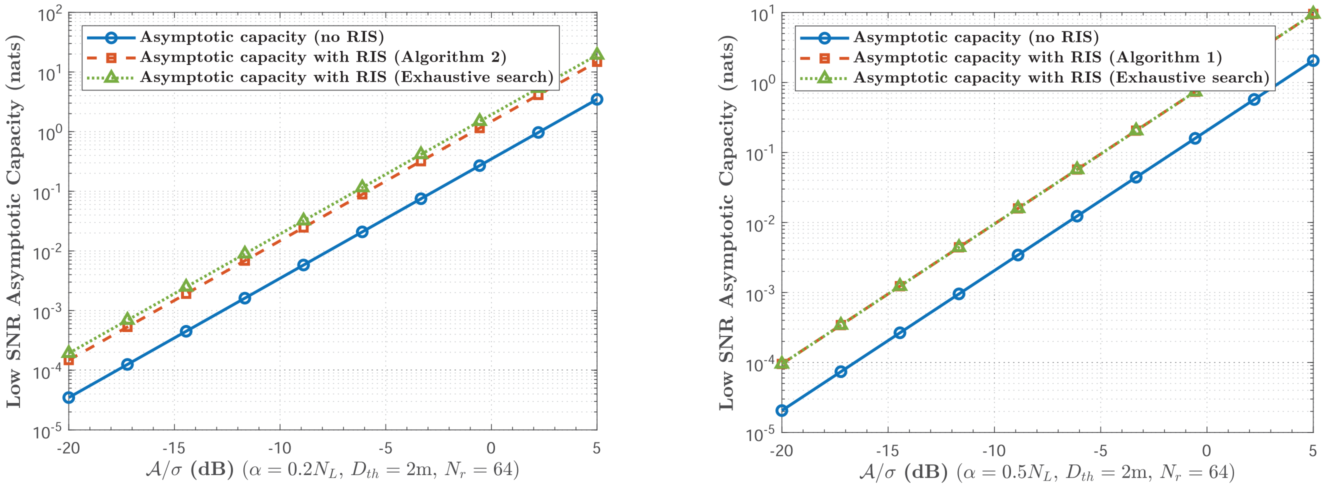
$$\mathrm{Tr}(\mathbf{H}) = \mathrm{Tr}(\boldsymbol{h}\boldsymbol{h}^\top) = \boldsymbol{h}^\top \boldsymbol{h} \leq \max \|\boldsymbol{h}\|^2. \tag{38}$$

Combined with the relaxed mapping relationship between $\mathbf{G}$ and $\mathbf{H}$, the optimization problem **P2-b** can be summarized as

$$\begin{aligned} \textit{P2-c:} \max_{\mathbf{H},\mathbf{G}} \quad & \mathrm{Tr}(\mathbf{QH}) \\ \text{s.t.} \quad & \begin{bmatrix} \mathbf{H} & \boldsymbol{h} \\ \boldsymbol{h}^\top & 1 \end{bmatrix} \succeq 0, \\ & \mathrm{Tr}(\mathbf{H}) \leq \chi, \\ & \boldsymbol{h} = \boldsymbol{h}^{\mathrm{LoS}} + \mathrm{diag}(\mathbf{H}^{\mathrm{R}})^\top \mathrm{vec}(\mathbf{G}), \\ & 0 \leq g_{n,i} \leq 1, \quad \forall n, i, \\ & \sum_{i=1}^{N_L} g_{n,i} \leq 1, \quad \forall n. \end{aligned} \tag{39}$$

where $\chi = \max \|\boldsymbol{h}\|^2$ and can be calculated by the constraints on assignment matrix.

Since $\mathbf{Q} \succeq 0$, the optimization problem **P2-c** constitutes a convex semidefinite program (SDP) problem that can be solved efficiently in polynomial time via interior-point methods or dedicated solvers such as CVX. While the original objective $\boldsymbol{h}^\top \mathbf{Q} \boldsymbol{h}$ in **P2-c** involves maximizing a convex function (which would generally render the problem non-convex), the reformulation of the objective as a linear function $\mathrm{Tr}(\mathbf{QH})$ over the auxiliary



(a) $\alpha \leq \frac{N_L}{2}$, $D_{\text{th}} = 2$m **(b)** $\alpha \geq \frac{N_L}{2}$, $D_{\text{th}} = 2$m

Fig. 2. Case I: Low SNR asymptotic capacity of the PSAI-constrained RIS-aided VLCP system when $\frac{\mathcal{A}}{\sigma} \to 0$ and $\epsilon = \alpha\mathcal{A}$.

matrix variable $\mathbf{H} = \boldsymbol{h}\boldsymbol{h}^\top$ resolves this conflict. Specifically, the convexity of **P2-c** is preserved by ensuring a linear objective and convex constraints. This structural transformation avoids the non-convexity inherent in maximizing quadratic forms by decoupling the quadratic dependence through the introduction of $\mathbf{H}$, thereby enabling guaranteed convergence to the global optimum. And the Gaussian randomization method is applied to deal with the rank 1 constraint $\text{rank}(\mathbf{H}) = 1$ to find the satisfied solution $\boldsymbol{h}^*$, and therefore, the derived relaxed solution $\tilde{\mathbf{G}}$ [33]. Details of the Gaussian randomization process can refer to Appendix B, and the mapping relation from $\boldsymbol{h}^*$ to $\tilde{\mathbf{G}}$ can refer to Appendix C.

Given the relaxed solution $\tilde{\mathbf{G}}$, the greedy thresholding method is introduced to complement the gap between the binary constraint and the relaxed operation. Compared with the branch-and-bound method, though it may only yield an approximate optimal solution, this method is still adopted in our work since the rapid convergence characteristic. Specifically, for each row n, the index i^* corresponding to the maximum value in the relaxed solution $\tilde{g}_{n,i}$ is selected, which can be formally expressed as

$$\begin{cases} g_{n,i^*} = 1, & i^* = \arg\max_i \tilde{g}_{n,i}, \\ g_{n,i} = 0, & i \neq i^*. \end{cases} \tag{40}$$

Consequently, the relaxed matrix $\tilde{\mathbf{G}}$ can be adjusted as $\mathbf{G}_{\text{comm}}$. Note that the optimal value for $\boldsymbol{\alpha}$ is related to $\boldsymbol{h}$, consequently, the optimal variable $\boldsymbol{\alpha}^*$ can be determined given a fixed $\boldsymbol{h}^{(t)}$ at the t-th iteration. Meanwhile, according to the optimal variable $\boldsymbol{\alpha}^*$, an updated $\boldsymbol{h}^{(t+1)}$ can be obtained at the $(t+1)$-th iteration. Consequently, the alternative optimization algorithm can be adopted to optimize $\boldsymbol{\alpha}$ and $\boldsymbol{h}$ alternatively until the convergence condition satisfy. Finally, the optimal $\mathbf{G}_{\text{comm}}$ can be determined after several iterations.

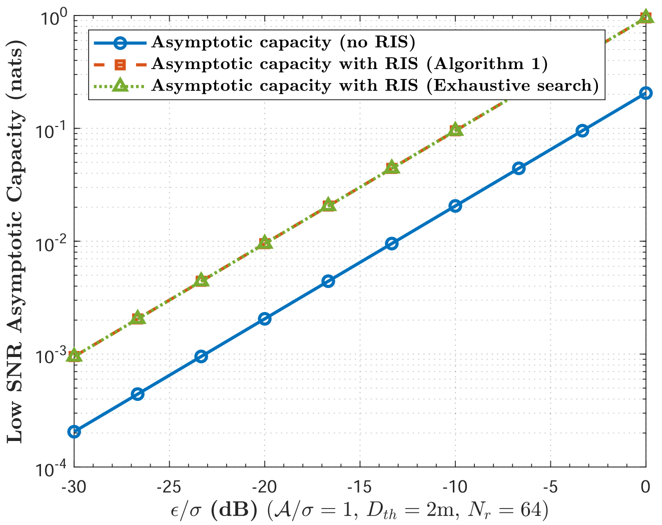
Combined with the greedy-based localization constraint satisfaction process and the convex relaxation-based communication

TABLE II
SIMULATION PARAMETERS

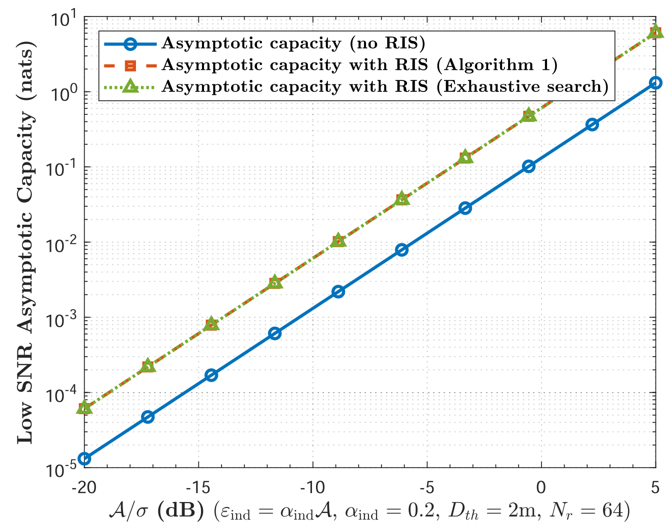
Parameter	Value
3D LED coordinates	$(2,2,3)$ $(2,6,3)$ $(6,2,3)$ $(6,6,3)$
LED half-angle $\Theta_{1/2}$	60°
User location	$(2.0, 3.2, 1.0)$ m
PD area A_{re}	0.01 m^2
PD refractive index β	1.5
FOV Ψ_{FoV}	80°
Filter gain T_{of}	1
Mirror reflectivity δ	0.9

performance maximization process, the optimal assignment matrix for **P2** can be obtained through configuration fusion. Details of the algorithm are summarized in Algorithm 2.

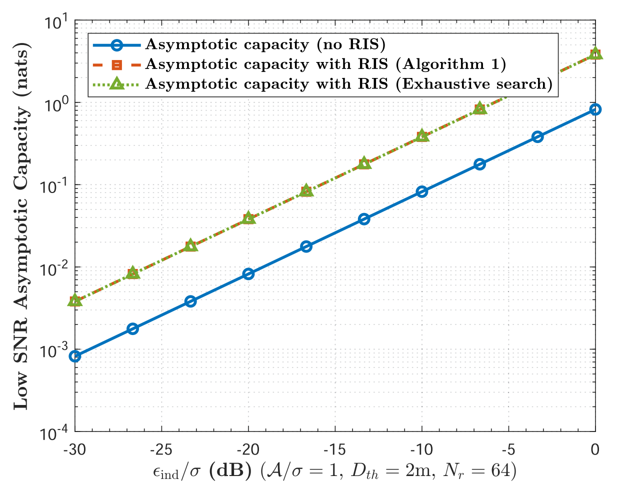
3) Baseline Algorithm for the Proposed Optimization Problems: For comparison, the exhaustive search algorithm is chosen as the baseline algorithm to demonstrate the efficiency of the proposed algorithms. The optimal solution for such complicating and difficult optimization problems can be obtained through the exhaustive search method. However, such a solution technique is highly intractable and cannot be realized in practice. In our work, the exhaustive search method is adopted to make sure the optimal solution of the proposed problem. In order to reduce the complexity, the satisfaction process for the location accuracy constraint is the same as that in Algorithm 1, and the optimization for the communication process is based on the exhaustive search algorithm in our work. Through traversing all the remaining mirrors of the RIS, the maximization for the communication performance can be achieved under the constraint on the



(a) Case II: Capacity for a PSAI-constrained RIS-aided VLCP system when $\frac{\epsilon}{\sigma} \to 0$ and $\frac{\mathcal{A}}{\sigma} = 1$.



(b) Case III: Capacity for a PIAI-constrained RIS-aided VLCP system when $\frac{\mathcal{A}}{\sigma} \to 0$ and $\epsilon_{\text{ind}} = \alpha_{\text{ind}}\mathcal{A}$.



(c) Case IV: Capacity for a PIAI-constrained RIS-aided VLCP system when $\frac{\epsilon_{\text{ind}}}{\sigma} \to 0$ and $\frac{\mathcal{A}}{\sigma} = 1$

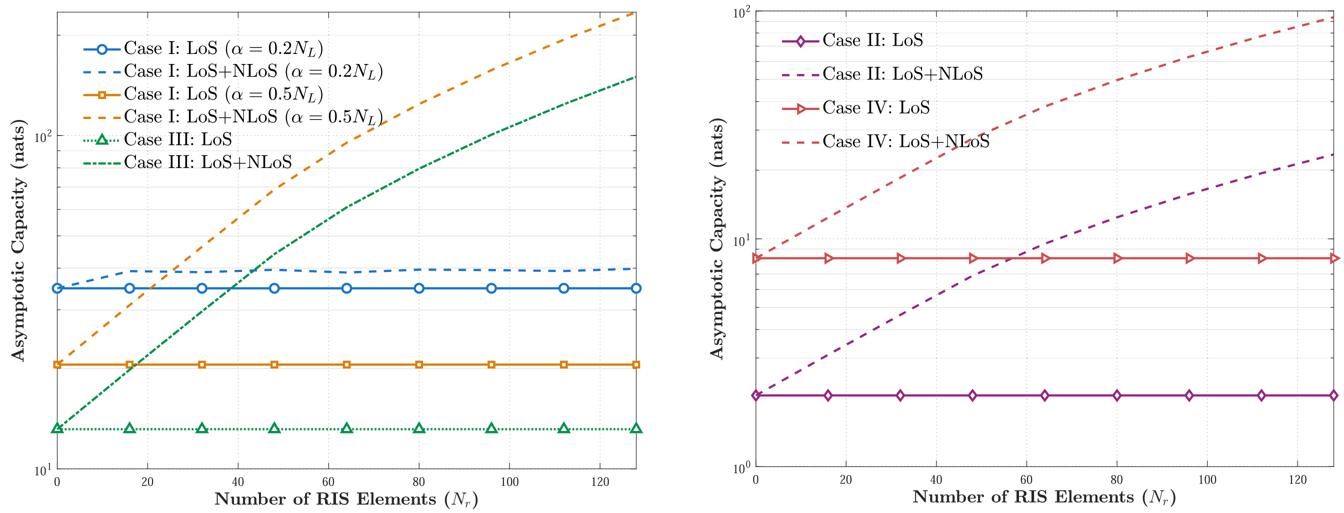
Fig. 3. Low SNR capacity results for the RIS-aided MISO VLCP system from Case II to Case IV.

positioning accuracy. Details of the exhaustive search algorithm are summarized in Algorithm 3.

C. Computational Complexity Analysis

This part discusses the computational complexity of the proposed optimization algorithm and baseline algorithm for the two proposed optimization problems. As for Algorithm 1, the computational complexity can be analyzed as follows. For the localization constraint satisfaction stage, the initialization of $\mathbf{G}_{\text{loc}}$ and FIM needs $\mathcal{O}(N_r N_L)$ operations. For the main while cycle, the worst-case number of iterations is $\mathcal{O}(N_r)$ (each mirror can be chosen at most once). For the inner cycle, the calculation of evaluable matrix $\mathbf{S}$ needs $\mathcal{O}(N_r N_L)$ operations, the chosen process of the optimal pair (n^*, i^*) needs $\mathcal{O}(N_r N_L)$ operations, the updating process of FIM and CRLB needs $\mathcal{O}(N_L^3)$ operations. For the communication process, the single cycle needs at most $\mathcal{O}(N_r N_L)$ operations, as it requires only a single pass over the matrix to identify the maximum element per row. Consequently, the worst-case overall computational complexity of Algorithm 1 is $\mathcal{O}(N_r^2 N_L + N_r N_L^3)$ overwhelmed by stage 1.

As for Algorithm 2, details of the computational complexity can be analyzed. As for the localization constraint satisfaction stage, the method is the same as that in Algorithm 1. Consequently, the complexity of this stage is $\mathcal{O}(N_r^2 N_L + N_r N_L^3)$. For the communication performance maximization process, considering the while loops of the alternative algorithm for optimizing $\mathbf{G}_{\text{comm}}$ and $\boldsymbol{\alpha}$, the complexity is dominated by the iterative process. Assuming the maximum iterations is denoted by $T_{\max}$, the complexity can be analyzed as follows. For each iteration, the complexity of the solution for the SDP



(a) Capacity for Case I and III versus the number of RIS elements. **(b)** Capacity for Case II and IV versus the number of RIS elements.

Fig. 4. Low SNR capacity for the RIS-aided VLCP system versus the number of RIS elements N_r.

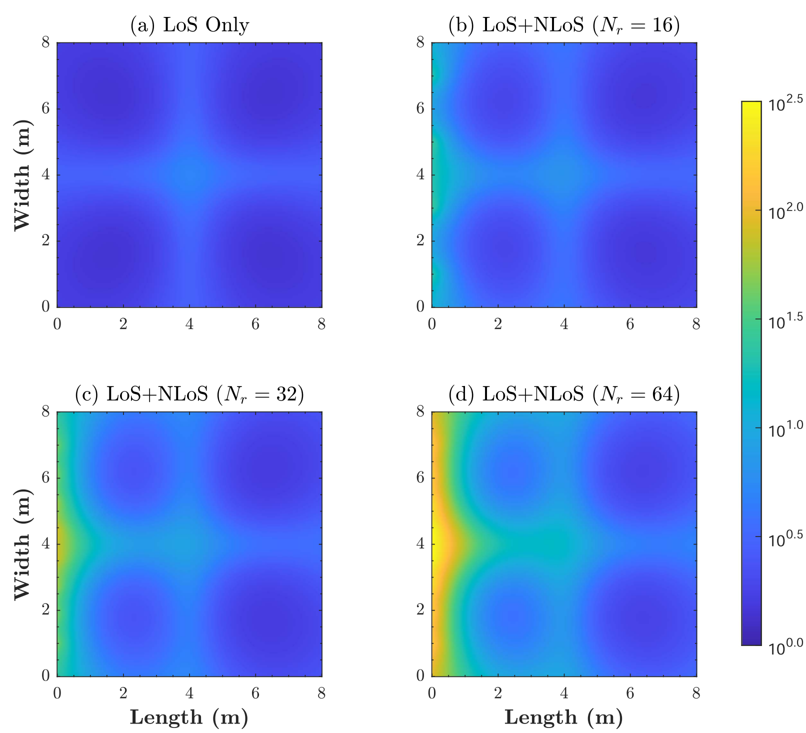
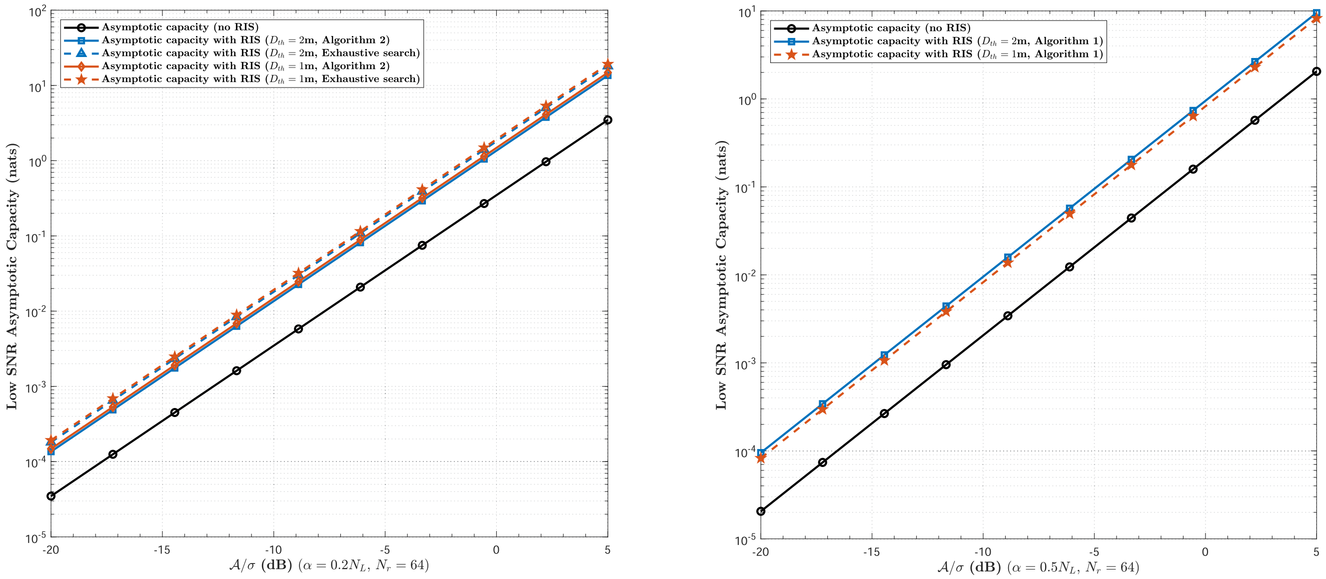


Fig. 5. Low SNR capacity for the RIS-aided VLCP system at different locations versus the number of RIS elements N_r.

problem **P2-c** is $\mathcal{O}((N_r N_L)^{3.5})$, which is the complexity for the typical SDP problem. The Gaussian randomization for generating candidate vector $\boldsymbol{h}^*$ needs $\mathcal{O}(T(N_r N_L)^2)$ operations per iteration, where T denotes the number of the randomization. The complexity of the matrix reconstruction process to obtain $\tilde{\mathbf{G}}$ is $\mathcal{O}((N_r N_L)^3\sqrt{N_r})$ per iteration analyzed via the inter-point method. Finally, the greedy strategy for the adjustment of $\tilde{\mathbf{G}}$ to obtain $\mathbf{G}_{\text{comm}}$ needs $\mathcal{O}(N_r N_L)$ operations per iteration. Combined with the two stages, the worst-case overall computational complexity of Algorithm 2 is $\mathcal{O}(T_{\max}((N_r N_L)^{3.5} + T(N_r N_L)^2 + ((N_r N_L)^3\sqrt{N_r})))$ overwhelmed by the solving process for the SDP problem.

As for the computational complexity analysis for the exhaustive search-based global configuration optimization algorithm, the complexity analysis can be divided into two parts. The complexity of the positioning optimization stage is $\mathcal{O}(N_r^2 N_L + N_r N_L^3)$. For the communication process, the exhaustive search algorithm for the RIS assignment matrix optimization problem



(a) $\alpha \leq \frac{N_L}{2}$

(b) $\alpha \geq \frac{N_L}{2}$

Fig. 6. Case I: Low SNR asymptotic capacity for the PSAI-constrained RIS-aided VLCP system versus different positioning accuracy constraints when $\frac{\mathcal{A}}{\sigma} \to 0$ and $\epsilon = \alpha\mathcal{A}$.

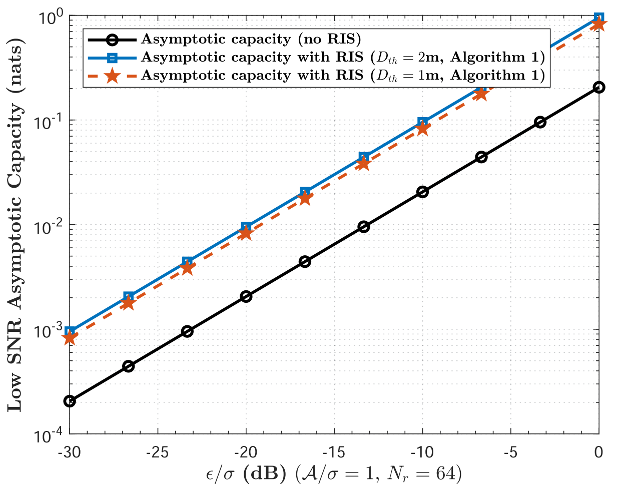
has a worst-case computational complexity of $\mathcal{O}(2^{N_r N_L})$ (there is no RIS element used for the location optimization process), which is significantly higher due to its brute-force nature. Consequently, the worst-case overall computational complexity of Algorithm 3 is $\mathcal{O}(N_r^2 N_L + N_r N_L^3) + \mathcal{O}(2^{N_r N_L}) \approx \mathcal{O}(2^{N_r N_L})$ overwhelmed by the exhaustive search process for the communication performance optimization. By comparing the complexity of Algorithm 3 and the proposed algorithms, we can observe that the proposed Algorithm 1 and Algorithm 2 can reach rapid convergence due to the low computational complexity. Through the comparison, it is obviously observed that the proposed low-complexity optimization algorithms are effective for the proposed optimization problems. Based on this, our proposed algorithms can achieve significant improvements in execution time and resource utilization. This is particularly important for real-time applications and large-scale RIS deployments where computational resources are limited.

V. Numerical Simulation Results

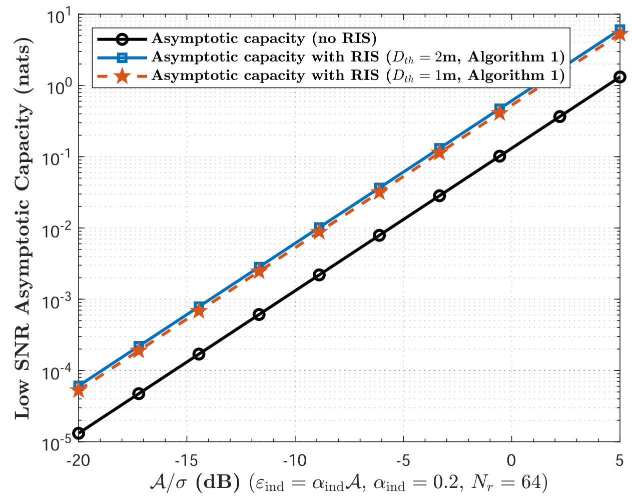
This section presents simulations that highlight the efficiency of our proposed algorithms in addressing the optimization problems formulated in the VLCP system with low SNR. Additionally, these simulations reveal the potential of specular reflecting RIS to enhance the performance of the VLCP system. Assuming that there are $N_L = 4$ LEDs and one user with one PD in an indoor room with size of $8\text{ m} \times 8\text{ m} \times 3\text{ m}$, the deployment of specular reflecting RIS with N_r elements is constrained in one rectangular region and centered on the center point of the wall. The rectangular region is determined by corners of $(0.0\text{ m}, 1.0\text{ m}, 1.2\text{ m})$ and $(0.0\text{ m}, 7.0\text{ m}, 2.9\text{ m})$ and the area of RIS element is assumed to be $5 \times 5\text{ m}^2$. The depth of the wall is ignored and the location of each RIS element can be determined according to the number of mirrors, as shown in Fig. 1. The receiver's FoV is assumed to be 80^{o} and the incident angles of LoS links or NLoS links are verified within the FoV. Most of the parameters are referred to [22], [34], [35] so as to ensure that our simulation environment is comparable to those used by other researchers. For ease of exposition, the LoS channel gain is normalized as $\|\boldsymbol{h}^{\text{LoS}}\|_2^2 = 1$, and meanwhile, the RIS-reflected channel vector $\boldsymbol{h}^{\text{NLoS}}$ and the Gaussian noise vector $\boldsymbol{z}$ are scaled by the same factor accordingly. All the other simulation parameters are summarized in Table. II.

Fig. 2 indicates the low SNR capacity results versus the peak SNR $\frac{\mathcal{A}}{\sigma}$ for the PSAI-constrained RIS-aided VLCP system with proportionally SNRs tends to zero, fixed positioning accuracy constraint $D_{\text{th}} = 2$ m and number of RIS elements $N_r = 64$. From the figure, "no RIS" means that the capacity is contributed by only LoS links. "Algorithm 1" and "Algorithm 2" denote that the assignment matrix is calculated by Algorithm 1 and Algorithm 2, respectively. "Exhaustive search" denotes that the optimal assignment matrix is determined by the exhaustive search algorithm. For the case, we consider the two types of capacity results with different values of α.

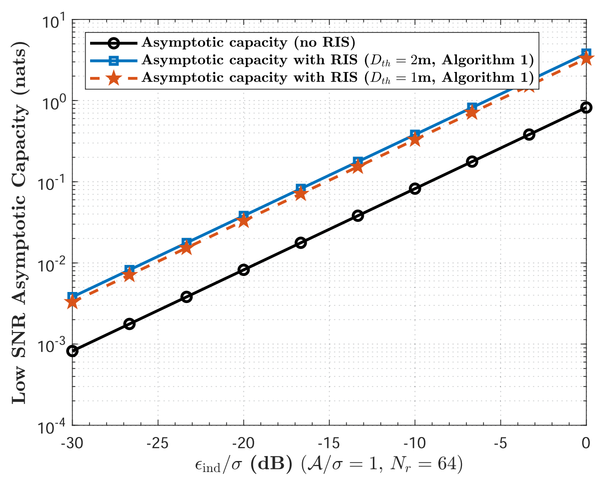
Fig. 2(a) represents the capacity results under the condition that $\alpha \leq \frac{N_L}{2}$, which can be calculated by Algorithm 2 and exhaustive search algorithm. The complexity of exhaustive search algorithm is extremely high and may reduce infeasible implement in actual practice. However, for ease of comparison, we consider the algorithm in our simulations with constrained number of mirrors $N_r = 64$ (more mirrors will need unacceptable computation time). From the figure, we can observe that the asymptotic capacity can be improved with the application of RIS, and meanwhile, the capacity results from the exhaustive search algorithm-based RIS assignment matrix is slightly higher



(a) Case II: Capacity with different positioning accuracy constraints for a PSAI-constrained VLCP system when $\frac{\epsilon}{\sigma}\to 0$ and $\frac{\mathcal{A}}{\sigma}=1$.



(b) Case III: Capacity with different positioning accuracy constraints for a PIAI-constrained VLCP system when $\frac{\mathcal{A}}{\sigma}\to 0$ and $\epsilon_{\text{ind}}=\alpha_{\text{ind}}\mathcal{A}$.



(c) Case IV: Capacity with different positioning accuracy constraints for a PIAI-constrained VLCP system when $\frac{\epsilon_{\text{ind}}}{\sigma}\to 0$ and $\frac{\mathcal{A}}{\sigma}=1$

Fig. 7. Low SNR capacity results for the RIS-aided MISO VLCP system versus different positioning accuracy constraints from Case II to Case IV.

than that from Algorithm 2. This is since the computational precision constraints of the SDP algorithm, as well as the relevant approximations in Algorithm 2. From Fig. 2(a), we can derive the efficiency of the proposed Algorithm 2 in solving the optimization problem.

Fig. 2(b) demonstrates the capacity results under the condition that $\alpha \geq \frac{N_L}{2}$, which can be calculated by Algorithm 1 and exhaustive search algorithm. From the figure, we can obtain that the capacity results calculated from Algorithm 1 is the same as that from the exhaustive search algorithm. This is since the monotonicity analyzed in **P1**. The efficiency of the Algorithm 1 in solving **P1** can be proved from the low complexity and good solution.

Fig. 3 demonstrates low SNR capacity results for three cases characterized by **P1**. The positioning accuracy constraint is fixed as $D_{\text{th}}=2$ m and the number of RIS elements is determined as $N_r=64$. The shared common term in the three cases is denoted by $\|\boldsymbol{h}\|_1^2$ and can be calculated by Algorithm 1. From the figure, we can observe the asymptotic capacity results for PSAI-constrained system with fixed peak SNR $\frac{\mathcal{A}}{\sigma}=1$ and vanished average SNR $\frac{\epsilon}{\sigma}\to 0$ (Fig. 3(a)), PIAI-constrained system with proportionally SNRs $\frac{\mathcal{A}}{\sigma}\to 0$ and $\epsilon_{\text{ind}}=\alpha_{\text{ind}}\mathcal{A}$ (Fig. 3(b)), PIAI-constrained system with fixed peak SNR $\frac{\mathcal{A}}{\sigma}=1$ and vanished individual average SNR $\frac{\epsilon_{\text{ind}}}{\sigma}\to 0$ (Fig. 3(c)). From the capacity results, we can obtain that the application of RIS can enhance the system performance of the VLCP system. Meanwhile, from the comparison between the Algorithm 1-based capacity results and Exhaustive search-based capacity results, it is proved that the proposed Algorithm 1 is an efficient method for the proposed RIS configuration problem.

Fig. 4 represents the simulation results of asymptotic capacity for RIS-aided VLCP system with different number of RIS elements. Specifically, the capacity results for proportionally SNRs are represented in Fig. 4(a) whereas that for fixed peak

SNR and vanished average SNR are demonstrated in Fig. 4(b). From the figure, we can observe that the increase of the number of RIS elements will promote the for VLCP system. Generally, more RIS elements will provide better capacity results and we can apply more mirrors in future to gain better performance. However, the increase on the capacity results in Case I is different from that in other cases. This is since that the change of $\boldsymbol{h}$ will change the value of $\mathbf{Q}$ in **P2-a**, and therefore, influence the capacity results. Consequently, for the special case, we should consider applying more mirrors carefully combined the potential performance increase and cost of RIS.

In Fig. 5, the low SNR asymptotic capacity for the RIS-aided VLCP system at different locations versus the number of RIS elements are demonstrated. From the figure, due to that the deployment of RIS is based on one wall in the simulated room, the capacity at locations which are close to the position of RIS is larger than that far away from the RIS. This is obvious since that the NLoS channel gain in (5) is better with short distance. Meanwhile, as the number of reflecting elements increases from $N_r = 16$ to $N_r = 64$, the capacity for the RIS-aided VLCP system is significantly enhanced. Concurrently, the area of the RIS-affected region also expands.

Figs. 6 and 7 indicate the influence of different positioning accuracy constraints on the asymptotic capacity results. Specifically, Fig. 6 demonstrates the influence of different positioning accuracy constraints on the capacity results for PSAI-constrained RIS-aided VLCP system at different α. Fig. 7 demonstrates the capacity results with different positioning accuracy constraints from Case II to Case IV. As for the capacity results with $\alpha \leq \frac{N_L}{2}$ in Case I (Fig. 6(a)), since the capacity results calculated by Algorithm 2 are different with that calculated by exhaustive search algorithm, we compare the two algorithm-based capacity results with $D_{\text{th}} = 2$ m and $D_{\text{th}} = 1$ m, respectively. As for the capacity results with $\alpha \geq \frac{N_L}{2}$ in Case I (Fig. 6(b)) and capacity results with different positioning accuracy constraints from Case II to Case IV (Fig. 7(a) to (c)), the exhaustive algorithm-based capacity results are ignored since that Algorithm 1 based capacity results with lower complexity are the same as the exhaustive algorithm-based capacity results.

From Figs. 6 and 7, we can obtain that the increase of the requirement on the positioning accuracy will deteriorate the communication performance. This is since that the higher requirement on the positioning accuracy will occupy more RIS elements with high derivative gain to decrease the CRLB, while remaining less RIS elements to maximize the communication performance determined by $\|\boldsymbol{h}\|_1$ or γ. Meanwhile, according to the capacity results in Fig. 6(a), we can observe that the influence of different algorithms on the capacity results cannot be ignored. In the future, we will develop more accurate and efficient optimization algorithms to address this type of capacity optimization problem. According to the simulations in Figs. 6 and 7, it is observed that the role of RIS in alleviating the contradiction between communication and positioning performance of VLCP systems can be demonstrated. Consequently, the proposed VLCP system with specular reflecting-based RIS has a significant advantage and plays an important role in future low SNR smart scenario such as biology darkroom lab or corridors.

VI. Conclusion

In this paper, we propose a specular reflecting-based RIS-aided VLCP system in low SNR and consider the design and optimization of the parameters of the RIS. Specifically, we analyze the low SNR asymptotic capacity for the system in four cases and CRLB for the positioning process. Meanwhile, we formulate two optimization problems according to the shared common term from the low SNR capacity results, with fixed positioning accuracy constraint. As for the two proposed optimization problems, combined with the convex relaxation, the monotonicity of the function, SDR and greedy algorithm, we propose the integrated two-stage greedy algorithm and greedy strategy-based convex relaxation algorithm to solve the optimization problems, respectively. Finally, we make some simulations to demonstrate the effectiveness of our proposed algorithms, and meanwhile, the performance increase caused by the application of RIS on the proposed VLCP system in low SNR can be demonstrated from our simulation results. Consequently, the proposed VLCP system, which incorporates a specular reflecting-based RIS, exhibits significant advantages and plays a crucial role in future low SNR smart scenario.

Appendix A
Proof of Proposition 1

Theorem 1: Let $\mathbf{Q} = \mathbf{M}(\boldsymbol{\alpha}) - \boldsymbol{\alpha}\boldsymbol{\alpha}^\top$ where $\mathbf{M}(\boldsymbol{\alpha})$ is a symmetric matrix with entries $M_{i,j} = \min\{\alpha_i, \alpha_j\}$, and $\boldsymbol{\alpha} \triangleq [\alpha_1, \ldots, \alpha_{N_L}]^\top$ with $\alpha_i \in [0,1], \forall i \in \mathcal{L}$. Then $\mathbf{Q}$ is positive semi-definite (PSD).

Proof: We construct a probabilistic interpretation and leverage covariance matrix properties to demonstrate the theorem. Let $U \sim \text{Uniform}[0,1]$ be a common random variable and define the Bernoulli random variables as

$$X_i = \mathbb{I}(U \leq \alpha_i) = \begin{cases} 1, & \text{if } U \leq \alpha_i, \\ 0, & \text{otherwise}, \end{cases} \tag{41}$$

where $\mathbb{I}(\cdot)$ denotes the indicator function, and consequently, the expectation and covariance matrix can be analyzed. Specifically, the first and second moments should satisfy

$$\mathbb{E}[X_i] = P(U \leq \alpha_i) = \alpha_i, \tag{42}$$

$$\mathbb{E}[X_i X_j] = P(U \leq \min\{\alpha_i, \alpha_j\}) = \min\{\alpha_i, \alpha_j\}. \tag{43}$$

Similarly, the covariance between X_i and X_j is derived by

$$\begin{aligned} \text{Cov}(X_i, X_j) &= \mathbb{E}[X_i X_j] - \mathbb{E}[X_i]\mathbb{E}[X_j] \\ &= \min\{\alpha_i, \alpha_j\} - \alpha_i \alpha_j \\ &= Q_{i,j}. \end{aligned} \tag{44}$$

Due to the reason that the covariance matrix of any random vector $\mathbf{X} = [X_1, \ldots, X_{N_L}]^\top$ is always PSD, and meanwhile, $\mathbf{Q}$ is exactly the covariance matrix of $\mathbf{X}$ (from (44)). Consequently, the following equation holds.

$$\mathbf{Q} \succeq 0. \tag{45}$$

In the proof process, the indicator function $\mathbb{I}(\cdot)$ ensures X_i is a Bernoulli random variable, the common random variable U

introduces dependence between X_i and X_j, the PSD property holds even if $\alpha_i = 0$ or 1 (boundary cases). Consequently, the theorem can be proved. □

Appendix B
Gaussian Randomization for Candidate Solution Generation

After obtaining the relaxed solution $\mathbf{H}^*$ via SDR, if $\mathbf{H}^*$ is not rank-1, the Gaussian randomization approach can be employed to generate candidate channel vectors $\boldsymbol{h}^*$ that satisfy the rank-1 constraint. These vectors are then mapped to the feasible solution $\tilde{\mathbf{G}}$. The detailed procedure consists of the following steps.

First of all, the eigenvalue decomposition is performed on $\mathbf{H}^*$ characterized by

$$\mathbf{H}^* = \sum_i \lambda_i \mathbf{v}_i \mathbf{v}_i^\top, \tag{46}$$

where λ_i and $\mathbf{v}_i$ denote the eigenvalues and corresponding eigenvectors. The dominant eigenvector $\mathbf{v}_1$ is extracted as a candidate direction. If the rank for $\mathbf{H}^*$ is 1, then the optimal channel vector can be obtained by $\boldsymbol{h}^* = \sqrt{\lambda_1}\mathbf{v}_1$. Otherwise, we generate L Gaussian random vectors $\boldsymbol{\xi}^{(\ell)}$ sampled from $\mathcal{N}(\mathbf{0}, \mathbf{H}^*)$ and quantize each vector to obtain candidate solutions based on

$$\boldsymbol{h}^{(\ell)} = \max\left(\boldsymbol{h}^{\text{LoS}},\ \text{ReLU}(\boldsymbol{\xi}^{(\ell)})\right), \quad \ell = 1, 2, \ldots, L, \tag{47}$$

where $\text{ReLU}(x) = \max(x, 0)$ guarantees the non-negativity and $\max(\cdot, \boldsymbol{h}^{\text{LoS}})$ ensures that the channel vector is no less than the LoS component. Finally, we implement the objective function evaluation and selection. The objective value for each candidate solution can be computed by

$$f\left(\boldsymbol{h}^{(\ell)}\right) = \boldsymbol{h}^{(\ell)\top}\mathbf{Q}\boldsymbol{h}^{(\ell)}. \tag{48}$$

The optimal candidate solution is selected based on

$$\boldsymbol{h}^* = \arg\max_{\boldsymbol{h}^{(\ell)}} f\left(\boldsymbol{h}^{(\ell)}\right). \tag{49}$$

This randomization method ensures feasibility while achieving near-optimal performance with sufficiently large L. The complexity scales linearly with the number of random samples L.

Appendix C
Mapping From Optimal Channel Vector to Relaxed Reflecting Matrix

Theorem 2: Let $\boldsymbol{h}^* \in \mathbb{R}_+^{N_L \times 1}$ be the optimized channel vector obtained through semi-definite relaxation, and $\mathbf{H}^{\text{R}} \in \mathbb{R}_+^{N_r \times N_L}$ the known reflection path gain matrix. There exists a relaxed reflecting matrix $\tilde{\mathbf{G}} \in [0,1]^{N_r \times N_L}$ satisfying

$$\boldsymbol{h}^{\text{NLoS}} = \boldsymbol{h}^* - \boldsymbol{h}^{\text{LoS}} = \left(\tilde{\mathbf{G}} \odot \mathbf{H}^{\text{R}}\right)^\top \mathbf{1}_{N_r}, \tag{50}$$

while adhering to the constraints

$$\sum_{i=1}^{N_L} \tilde{g}_{n,i} \leq 1, \quad \forall n \in \{1, \ldots, N_r\}, \tag{51}$$

$$0 \leq \tilde{g}_{n,i} \leq 1, \quad \forall n, i. \tag{52}$$

Proof: The mapping procedure comprises two key steps. Firstly, the NLoS channel vector can be derived by

$$\boldsymbol{h}^{\text{NLoS}} = \boldsymbol{h}^* - \boldsymbol{h}^{\text{LoS}}, \tag{53}$$

where $\boldsymbol{h}^{\text{LoS}}$ is the deterministic LoS channel vector. Secondly, the constrained least-squares formulation is expressed as

$$\min_{\tilde{\mathbf{G}}} \left\| \boldsymbol{h}^{\text{NLoS}} - \left(\tilde{\mathbf{G}} \odot \mathbf{H}^{\text{R}}\right)^\top \mathbf{1}_{N_r} \right\|_2^2$$
$$\text{s.t.} \quad \sum_{i=1}^{N_L} \tilde{g}_{n,i} \leq 1, \quad 0 \leq \tilde{g}_{n,i} \leq 1. \tag{54}$$

This convex problem can be efficiently solved using CVX or interior-point methods.

Thus, $\tilde{\mathbf{G}}$ provides a feasible relaxed solution that bridges the optimal channel vector $\boldsymbol{h}^*$ and the discrete reflecting matrix $\mathbf{G}$. □

Disclosures

The authors declare no conflicts of interest.

Acknowledgment

The authors wish to thank the anonymous reviewers for their valuable suggestions.

References

[1] C. Zeyu, “6G, LIFI and WIFI wireless systems: Challenges, development and prospects,” in *Proc. 18th Int. Comput. Conf. Wavelet Act. Media Technol. Inf. Process.*, 2021, pp. 322–325.

[2] G. Zhu et al., “Pushing AI to wireless network edge: An overview on integrated sensing, communication, and computation towards 6G,” *Sci. China Inf. Sci.s*, vol. 66, no. 3, 2023, Art. no. 130301.

[3] S. Ma et al., “Optimal power allocation for integrated visible light positioning and communication system with a single LED-lamp,” *IEEE Trans. Commun.*, vol. 70, no. 10, pp. 6734–6747, Oct. 2022.

[4] Y. Zhang, S. Kasahara, Y. Shen, X. Jiang, and J. Wan, “Smart contract-based access control for the Internet of Things,” *IEEE Internet Things J.*, vol. 6, no. 2, pp. 1594–1605, Apr. 2019.

[5] D. Tsonev, S. Videv, and H. Haas, “Towards a 100 Gb/s visible light wireless access network,” *Opt. Exp.*, vol. 23, no. 2, pp. 1627–1637, 2015.

[6] L. Shi, B. Béchadergue, L. Chassagne, and H. Guan, “Joint visible light sensing and communication using m-Cap modulation,” *IEEE Trans. Broadcast.*, vol. 69, no. 1, pp. 276–288, Mar. 2023.

[7] H. Yang, C. Chen, W.-D. Zhong, A. Alphones, S. Zhang, and P. Du, “Demonstration of a quasi-gapless integrated visible light communication and positioning system,” *IEEE Photon. Technol. Lett.*, vol. 30, no. 23, pp. 2001–2004, Dec. 2018.

[8] M. Nassiri, G. Baghersalimi, and Z. Ghassemlooy, “A hybrid VLP and VLC system using m-Cap modulation and fingerprinting algorithm,” *Opt. Commun.*, vol. 473, 2020, Art. no. 125699.

[9] D. Chen et al., “Integrated visible light communication and positioning CDMA system employing modified ZCZ and walsh code,” *Opt. Exp.*, vol. 30, no. 22, pp. 40455–40469, 2022.

[10] H. Yang, W.-D. Zhong, C. Chen, A. Alphones, and P. Du, “QoS-driven optimized design-based integrated visible light communication and positioning for indoor IoT networks,” *IEEE Internet Things J.*, vol. 7, no. 1, pp. 269–283, Jan. 2020.

[11] H. Yang et al., “Coordinated resource allocation-based integrated visible light communication and positioning systems for indoor IoT,” *IEEE Trans. Wireless Commun.*, vol. 19, no. 7, pp. 4671–4684, Jul. 2020.

[12] S. Ma et al., “Waveform design and optimization for integrated visible light positioning and communication,” *IEEE Trans. Commun.*, vol. 71, no. 9, pp. 5392–5407, Sep. 2023.

[13] S. Ma et al., "Robust power allocation for integrated visible light positioning and communication networks," *IEEE Trans. Commun.*, vol. 71, no. 8, pp. 4764–4777, Aug. 2023.

[14] L. Yin, W. O. Popoola, X. Wu, and H. Haas, "Performance evaluation of non-orthogonal multiple access in visible light communication," *IEEE Trans. Commun.*, vol. 64, no. 12, pp. 5162–5175, Dec. 2016.

[15] S. Aboagye, A. R. Ndjiongue, T. M. Ngatched, O. A. Dobre, and H. V. Poor, "Ris-Assisted visible light communication systems: A tutorial," *IEEE Commun. Surveys Tuts.*, vol. 25, no. 1, pp. 251–288, Firstquarter 2023.

[16] S. Aboagye, T. M. Ngatched, O. A. Dobre, and A. R. Ndjiongue, "Intelligent reflecting surface-aided indoor visible light communication systems," *IEEE Commun. Lett.*, vol. 25, no. 12, pp. 3913–3917, Dec. 2021.

[17] S. Sun, F. Yang, J. Song, and Z. Han, "Joint resource management for intelligent reflecting surface–aided visible light communications," *IEEE Trans. Wireless Commun.*, vol. 21, no. 8, pp. 6508–6522, Aug. 2022.

[18] S. Sun, F. Yang, J. Song, and Z. Han, "Optimization on multiuser physical layer security of intelligent reflecting surface-aided VLC," *IEEE Wireless Commun. Lett.*, vol. 11, no. 7, pp. 1344–1348, Jul. 2022.

[19] S. Xu, Y. Wu, L. Zhang, S. Xing, and F. Wei, "Simultaneous position and orientation estimation for reconfigurable intelligent surfaces-assisted visible light communications," *IEEE Trans. Veh. Technol.*, vol. 74, no. 1, pp. 861–876, Jan. 2025.

[20] F. Kokdogan and S. Gezici, "Intelligent reflecting surfaces for visible light positioning based on received power measurements," *IEEE Trans. Veh. Technol.*, vol. 73, no. 9, pp. 13108–13121, Sep. 2024.

[21] Q. Wu, J. Zhang, Y.-y. Zhang, G. Xin, and D. Li, "Reconfigurable intelligent surface-aided positioning-centered multi-LED integrated visible light communication and positioning system," *Opt. Exp.*, vol. 33, no. 3, pp. 5227–5252, 2025.

[22] A. Chaaban, Z. Rezki, and M.-S. Alouini, "Low-SNR asymptotic capacity of MIMO optical intensity channels with peak and average constraints," *IEEE Trans. Commun.*, vol. 66, no. 10, pp. 4694–4705, Oct. 2018.

[23] S. Abdeljabar, M. W. Eltokhey, and M.-S. Alouini, "Sum rate and fairness optimization in RIS-assisted VLC systems," *IEEE Open J. Commun. Soc.*, vol. 5, pp. 2555–2566, 2024.

[24] T. Komine and M. Nakagawa, "fundamental analysis for visible-light communication system using LED lights," *IEEE Trans. Consum. Electron.*, vol. 50, no. 1, pp. 100–107, Feb. 2004.

[25] S. Sun, W. Mei, F. Yang, N. An, J. Song, and R. Zhang, "Optical intelligent reflecting surface assisted MIMO VLC: Channel modeling and capacity characterization," *IEEE Trans. Wireless Commun.*, vol. 23, no. 3, pp. 2125–2139, Mar. 2024.

[26] A. M. Abdelhady, A. K. S. Salem, O. Amin, B. Shihada, and M.-S. Alouini, "Visible light communications via intelligent reflecting surfaces: Metasurfaces vs mirror arrays," *IEEE Open J. Commun. Soc.*, vol. 2, pp. 1–20, 2021.

[27] S. Sun, F. Yang, W. Mei, J. Song, Z. Han, and R. Zhang, "Optical IRS for visible light communication: Modeling, design, and open issues," *IEEE Wireless Commun.*, vol. 32, no. 4, pp. 162–169, 2025.

[28] S. M. Moser, L. Wang, and M. Wigger, "Capacity results on multiple-input single-output wireless optical channels," *IEEE Trans. Inf. Theory*, vol. 64, no. 11, pp. 6954–6966, Nov. 2018.

[29] M. F. Keskin, S. Gezici, and O. Arikan, "Direct and two-step positioning in visible light systems," *IEEE Trans. Commun.*, vol. 66, no. 1, pp. 239–254, Jan. 2018.

[30] H. V. Poor, *An Introduction to Signal Detection and Estimation*. Berlin, Germany: Springer, 2013.

[31] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, *Introduction to Algorithms*. Cambridge, MA, USA: MIT Press, 2022.

[32] C. Lopez, *MATLAB Optimization Techniques*. New York, NY, USA: Apress, 2014.

[33] Z.-q. Luo, W.-k. Ma, A. M.-c. So, Y. Ye, and S. Zhang, "Semidefinite relaxation of quadratic optimization problems," *IEEE Signal Process. Mag.*, vol. 27, no. 3, pp. 20–34, May 2010.

[34] S. Sun, N. An, F. Yang, J. Song, and Z. Han, "Capacity characterization analysis of optical intelligent reflecting surface assisted MISO VLC," *IEEE Internet Things J.*, vol. 11, no. 3, pp. 4801–4814, Feb. 2024.

[35] S. Sun, F. Yang, and J. Song, "Sum rate maximization for intelligent reflecting surface-aided visible light communications," *IEEE Commun. Lett.*, vol. 25, no. 11, pp. 3619–3623, Nov. 2021.